

Deep Learning Framework for Composite Material Analysis and Damage Assessment in CFRP Structures

B.Tech (AIDS)
UG-II (IV-Semester)

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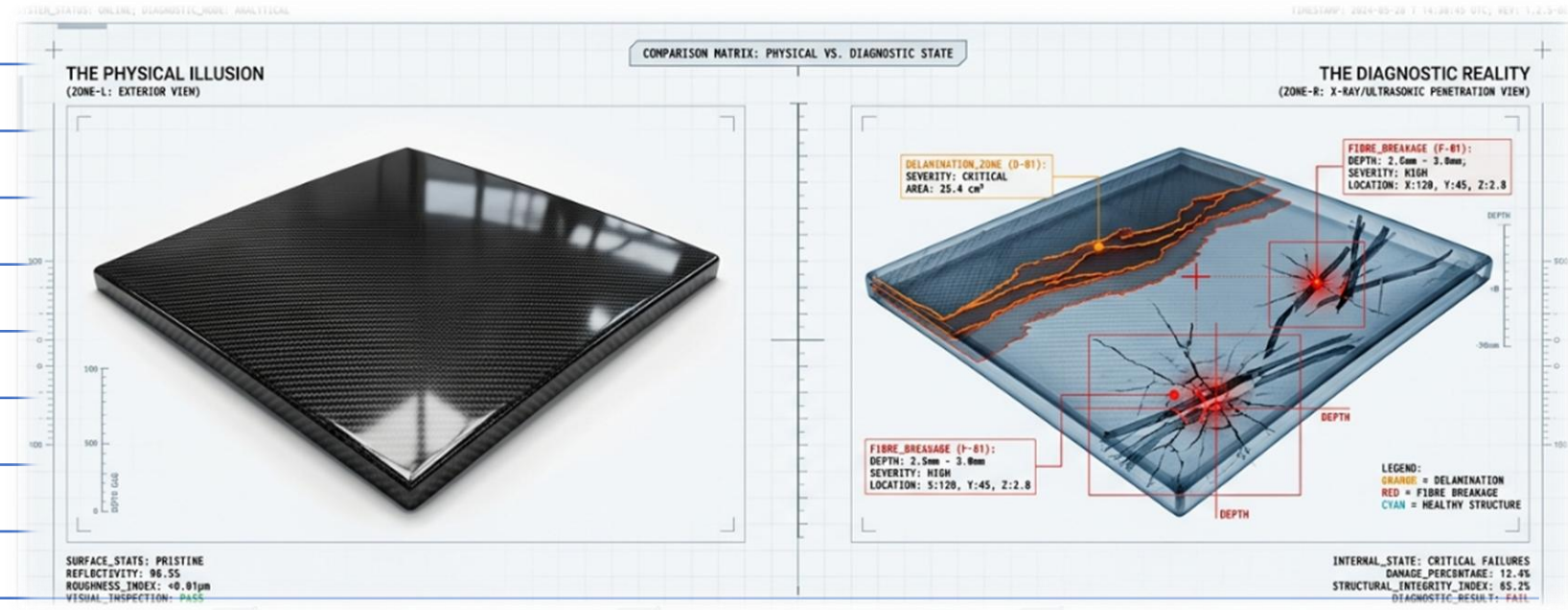


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UNIFIED MULTIMODAL EXPLAINABLE DEEP LEARNING FRAMEWORK FOR COMPOSITE MATERIAL ANALYSIS & DAMAGE ASSESSMENT IN CFRP STRUCTURES

STAGE 1

DATA ACQUISITION & PREPROCESSING

TABULAR MATERIAL PARAMETERS

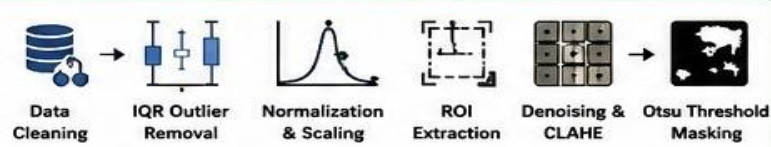
- Fibre Type
- Resin Type
- Density (ρ)
- Void Content (V_v)
- Fibre Volume Fraction (V_f)
- Cure Temperature
- Crack Length (a)
- Defect Density ...

Sample	1	2	3	4	5	6	7	8	9	10
Material	Carbon	Carbon	Carbon	Carbon	Carbon	Carbon	Carbon	Carbon	Carbon	Carbon
Resin	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy	Epoxy
Density	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Void Content	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Fibre Volume Fraction	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Cure Temperature	120	120	120	120	120	120	120	120	120	120
Crack Length	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Defect Density	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01

CFRP IMAGE DATA



PREPROCESSING PIPELINE



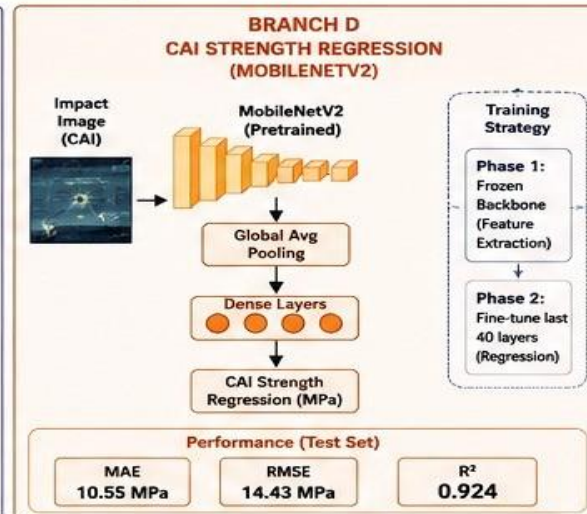
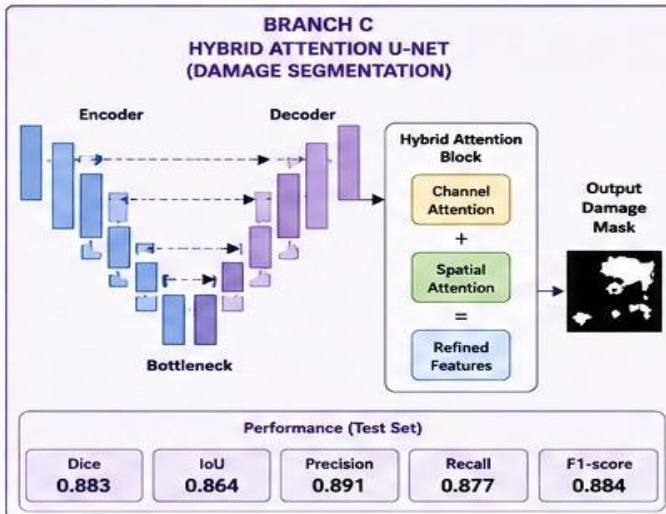
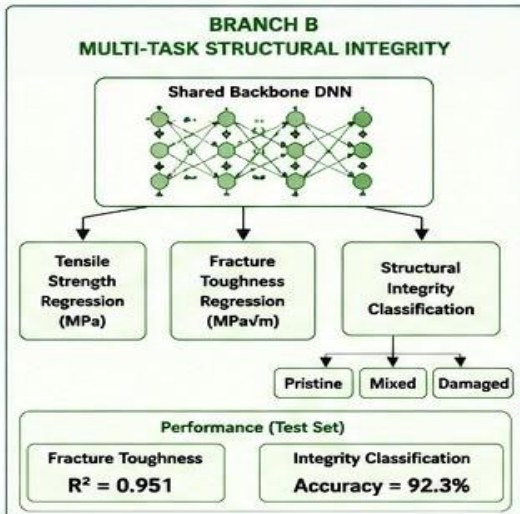
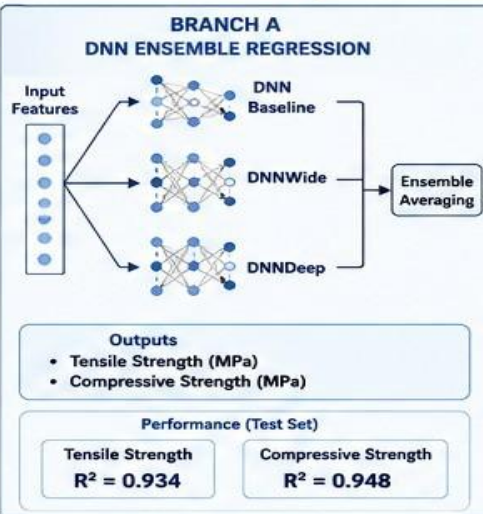
PHYSICS-INFORMED FEATURE ENGINEERING

$$\sigma_c = \sigma_t \times V_f \times K_\mu \times K_f \times K_r \times \hat{\rho} \times (1 - 0.05V_v)$$

- Incorporates material mechanics knowledge
- Improves generalization & physical consistency
- Generates compressive strength target

STAGE 2

MULTI-BRANCH DEEP LEARNING ARCHITECTURE



STAGE 3

TRAINING & OPTIMIZATION

Loss Functions

- MSE / Huber Loss (Regression)
- Cross-Entropy Loss (Classification)



Optimizer

Adam (lr = 1e-3)



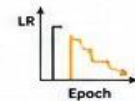
Regularization

- Batch Normalization
- Dropout (0.2 - 0.3)
- L2 Weight Decay



Training Strategy

- Early Stopping
- ReduceLROnPlateau
- Fine-tuning



Data Split

Train : 70% (280)
Validation : 15% (60)
Test : 15% (60)



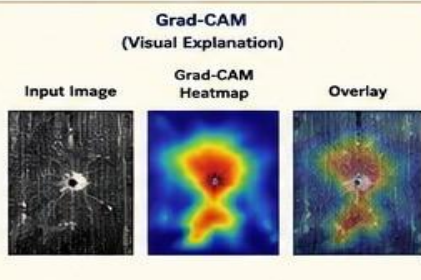
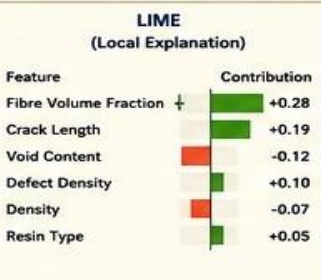
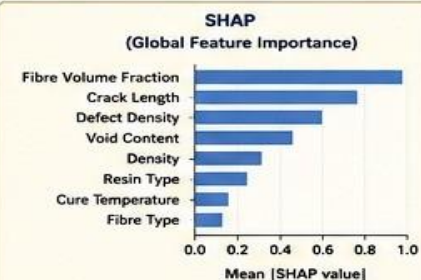
Hardware & Framework



GPU Acceleration

STAGE 4

EXPLAINABILITY & INSIGHTS



FINAL ENGINEERING OUTPUTS & IMPACT

- ✓ Accurate Strength Prediction (Tensile, Compressive, CAI)
- ✓ Structural Integrity Classification
- ✓ Pixel-Level Damage Segmentation
- ✓ Physically Informed & Explainable AI
- ✓ Low-Cost, Scalable & Accessible NDE
- ✓ Supports Design, Certification & SHM

OVERALL FRAMEWORK PERFORMANCE (TEST SET SUMMARY)

Regression Tasks (Best R ²)	0.948
Integrity Classification Accuracy	92.3%
Damage Segmentation (Dice)	0.883
CAI Strength Regression (R ²)	0.924

Consistent High Performance Across All Modalities



Problem Statement

Engineering Problem: CFRP composites used in aerospace and automotive industries develop hidden damages such as cracks, delamination, and fiber breakage that are difficult to detect manually and may lead to structural failure.

Importance: Traditional inspection techniques like ultrasonic testing and X-ray CT scanning are expensive, slow, and require specialized equipment, making continuous monitoring challenging.

Real-World Applications: Automated damage detection and strength prediction are important for aircraft structures, wind turbine blades, automobiles, bridges, and defense systems to improve safety and reliability.

Need for Intelligent System: An explainable deep learning framework is required for accurate, fast, low-cost, and automated structural health monitoring and damage assessment of CFRP materials.



Literature Review

Paper 1: “Deep learning-based compression after impact strength prediction of CFRP laminates using thermographic images”

Year: 2021

Journal: *Composite Structures*

Solution: Predicted Compression After Impact (CAI) strength of CFRP laminates using CNN models and thermographic images.

Methodology: Used thermographic imaging with deep convolutional neural networks for strength regression and defect analysis.

Limitation: Required expensive thermal imaging equipment and focused only on CAI prediction.

Our Improvement: Our work uses standard optical images instead of thermography and provides a unified multimodal framework including regression, classification, segmentation, and explainable AI.

Paper2: “Deep learning for defect detection in composite materials using convolutional neural networks on ultrasonic C-scan images”

Year: 2020

Journal: *NDT & E International*

Solution: Automated defect detection in composite materials using CNN-based image analysis.

Methodology: Applied convolutional neural networks on ultrasonic C-scan images for defect classification and localization.

Limitation: Relied only on ultrasonic data and lacked interpretability and multimodal learning.

Our Improvement: Our framework integrates SHAP, LIME, and Grad-CAM explainability with hybrid attention U-Net and supports multiple CFRP analysis tasks.



Literature Review

Paper 3(Amrita): “Experimental validation of compressive strength prediction using machine learning algorithm”

Year: 2022

Journal: *Materials Today: Proceedings*

Solution: Predicted compressive strength of fiber reinforced concrete using ML algorithms.

Methodology: Used Random Forest, ANN, SVR, and regression models on experimental material datasets.

Limitation: Focused only on compressive strength prediction and lacked image-based analysis.

Our Improvement: Our work performs multimodal analysis including image segmentation, structural integrity classification, and explainable deep learning for CFRP composites.

Paper 4(Amrita): “Machine learning driven advances in hardness prediction and artificial intelligence based phase classification in aluminium-high entropy alloy systems”

Year: 2026

Journal: *Journal of Alloys and Compounds*

Solution: Predicted hardness and phase classification in alloy systems using ML and AI models.

Methodology: Used CatBoost, XGBoost, ANN, and SHAP-based explainability for material property prediction.

Limitation: Limited to tabular material datasets and did not include structural damage localization or segmentation.

Our Improvement: Our framework combines tabular and image-based learning with hybrid attention U-Net for accurate damage assessment and CAI prediction in CFRP structures.



Objective of the Project

1. Develop Unified Framework: Design a multimodal deep learning framework for composite material analysis using both tabular and image-based data.

2. Predict Material Properties: Perform tensile strength, compressive strength, and fracture toughness prediction with high accuracy.

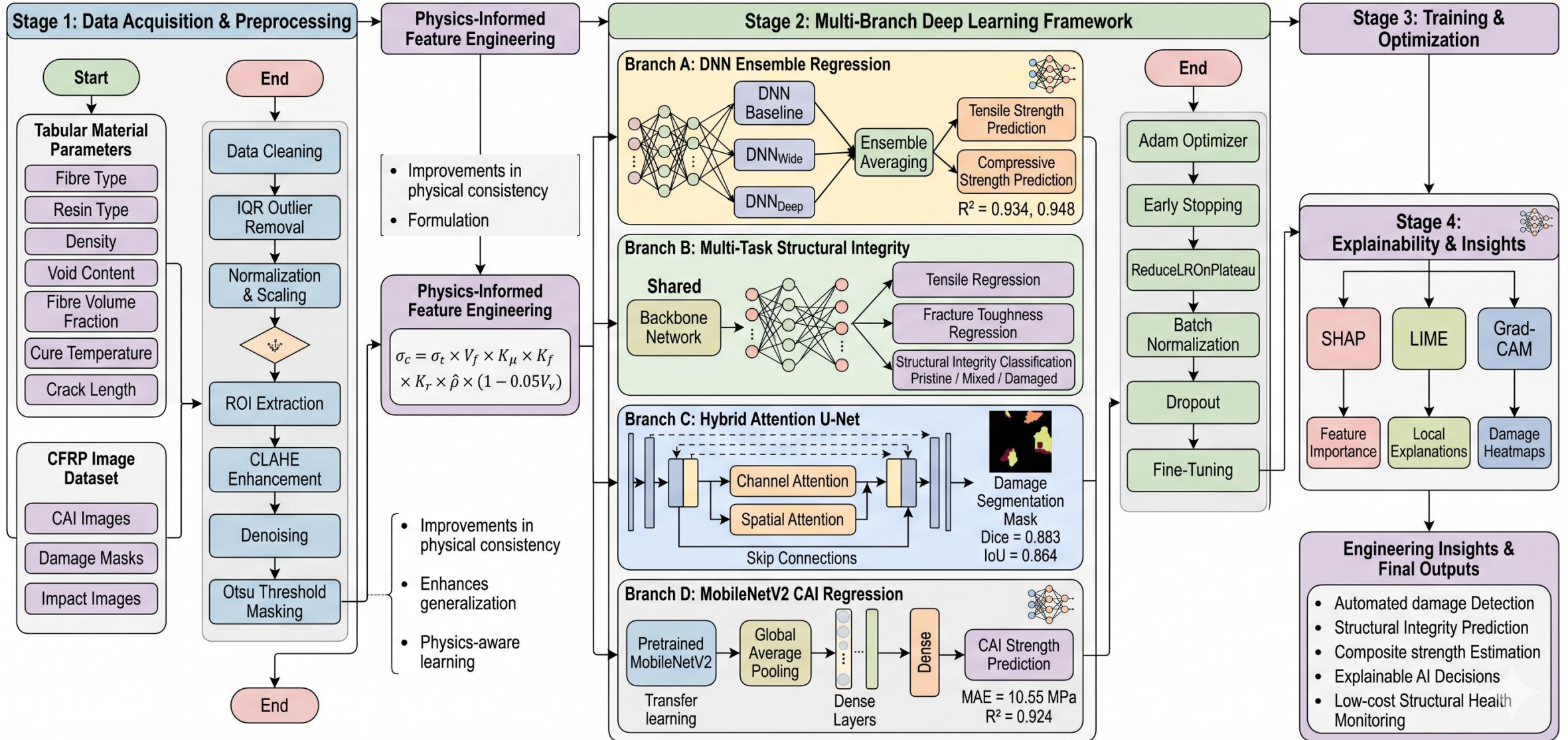
3. Detect Structural Damage: Develop a Hybrid Attention U-Net model for accurate pixel-level damage segmentation in CFRP structures.

4. Perform CAI Strength Regression: Predict Compression-After-Impact (CAI) strength directly from CFRP specimen images using transfer learning.

5. Integrate Explainable AI: Use SHAP, LIME, and Grad-CAM to improve interpretability and visualize model decision-making.

6. Improve Structural Health Monitoring: Enable automated, low-cost, and reliable monitoring of composite structures for real-world engineering applications.

Proposed Methodology

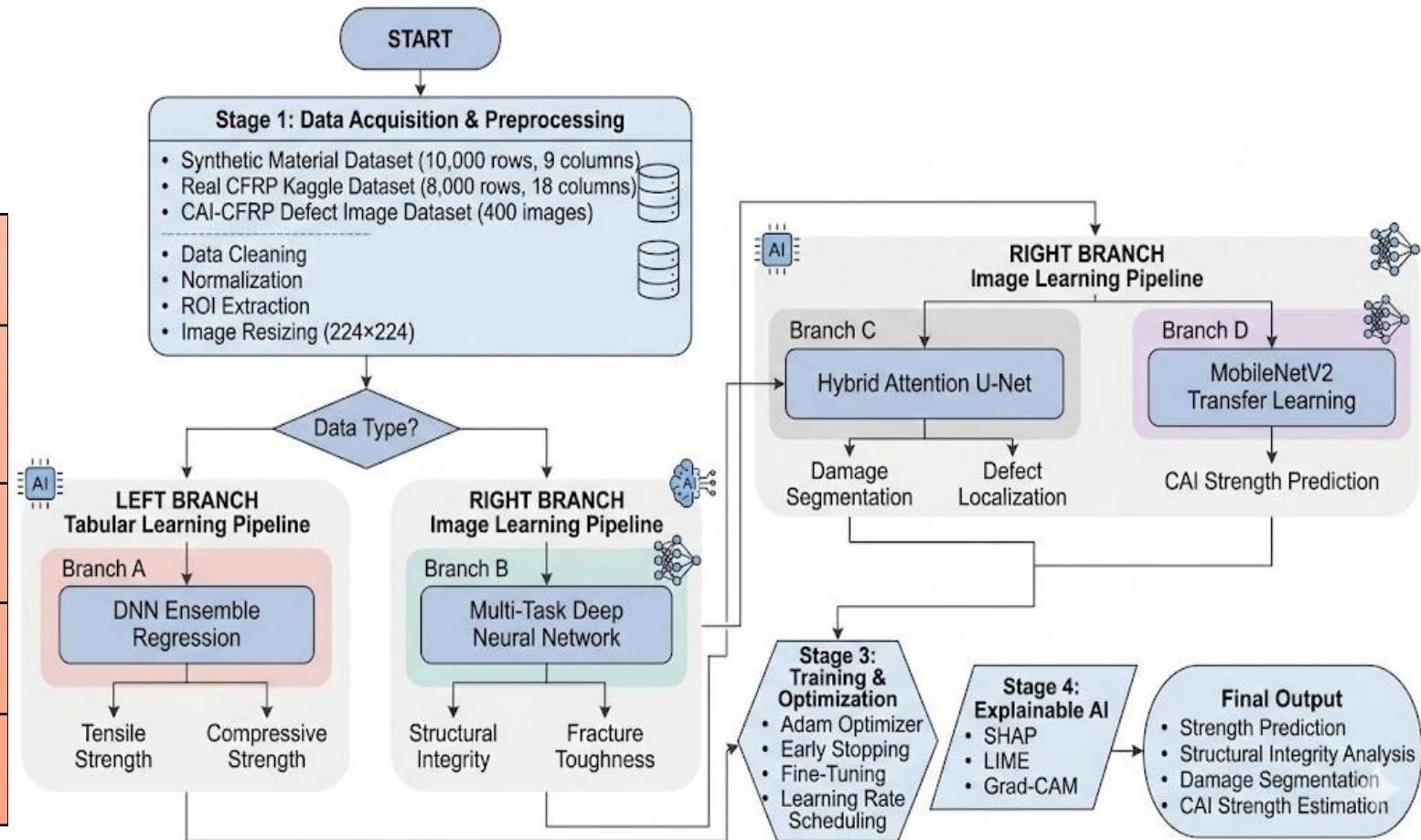


Proposed Methodology

It combines synthetic and real CFRP material data, defect images, physics-based learning, XAI, and DL models to perform key tasks such as strength prediction, fracture analysis, damage detection, and CAI strength estimation in CFRP structures.

It consists of 4 parallel AI branches:

Branch	Purpose	Model
A	Tensile & Compressive Strength Prediction	DNN Ensemble
B	Structural Integrity & Fracture Toughness	Multi-Task DNN
C	Damage Segmentation	Hybrid Attention U-Net
D	CAI Strength Prediction	MobileNetV2





Unified Multimodal Explainable Deep Learning Framework

Stage	Branch	Purpose	Techniques
Stage 1	Data Acquisition & Preprocessing	Prepare high-quality multimodal datasets for training	Collection of tabular and image datasets ,Data cleaning and normalization and ROI extraction and image enhancement
Stage 2	Branch A — Strength Prediction	Predict tensile and compressive strength of CFRP composites	DNN Ensemble Regression
	Branch B — Structural Integrity Analysis	Estimate fracture toughness and structural integrity	Multi-Task DNN Learning
	Branch C — Damage Segmentation	Detect and segment internal damage regions	Hybrid Attention U-Net
	Branch D — CAI Strength Prediction	Predict Compression-After-Impact (CAI) strength from images	MobileNetV2 Transfer Learning
Stage 3	Training & Optimization	Improve convergence, stability, and model performance	• Adam Optimizer • Early Stopping • Fine-Tuning • Learning-Rate Scheduling
Stage 4	Explainability & Insights	Provide interpretable and trustworthy AI predictions	SHAP , LIME and Grad-CAM Visualization



Data Acquisition

Dataset	Dataset Type & Source	Data Description	Input Features / Images	Target Outputs	Associated Model / Task
Dataset 1	Material Dataset(Kaggle)	Artificially generated tabular dataset used for material-property learning and strength prediction	• Fibre type • Resin type • Density • Void content • Cure temperature • Fibre volume fraction • Crack length • Additional material parameters Size: ~10,000 rows $\times$ 9 columns	• Tensile strength • Compressive strength(generated)	DNN Ensemble Regression for strength prediction
Dataset 2	Real CFRP Kaggle Dataset (CSV)	Real-world CFRP composite dataset collected from Kaggle for structural integrity and fracture analysis	• Fibre orientation • Defect density • Crack length • Void content • Multiple composite parameters Size: ~8,000 rows $\times$ 18 columns	• Structural integrity class • Fracture toughness	Multi-Task Deep Neural Network for structural integrity assessment
Dataset 3	CAI-CFRP Defect Image Dataset(Kaggle)	CFRP post-impact image dataset used for damage localization, segmentation, and CAI prediction	• CFRP defect images • Binary segmentation masks • Images resized to 224 $\times$ 224 Original Images: ~800 Used Images: 400 (selected due to computational complexity)	• Damage segmentation maps • CAI strength labels	Hybrid Attention U-Net for segmentation and MobileNetV2 Transfer Learning for CAI strength estimation



Physics-Informed Feature Engineering & Branch A

Physics-Informed Learning

Aspect	Details
Objective	Integrates engineering mechanics with AI models
Purpose	Improve physical consistency and realistic prediction behavior
Key Idea	Material science equations guide feature learning and prediction constraints

Compressive Strength Equation

$$\sigma_c = \sigma_t \cdot V_f \cdot K_\mu \cdot K_f \cdot K_r \cdot \hat{\rho} \cdot (1 - 0.05V_v)$$

Symbol	Meaning
σ_c	Compressive strength
σ_t	Tensile strength
V_f	Fibre volume fraction
K_μ	Fibre orientation factor
K_f	Fibre factor
K_r	Resin factor
$\hat{\rho}$	Density parameter
V_v	Void volume fraction

Physics-informed learning reduces unrealistic predictions, improves generalization, incorporates engineering knowledge, and enhances prediction reliability.

Branch A — DNN Ensemble

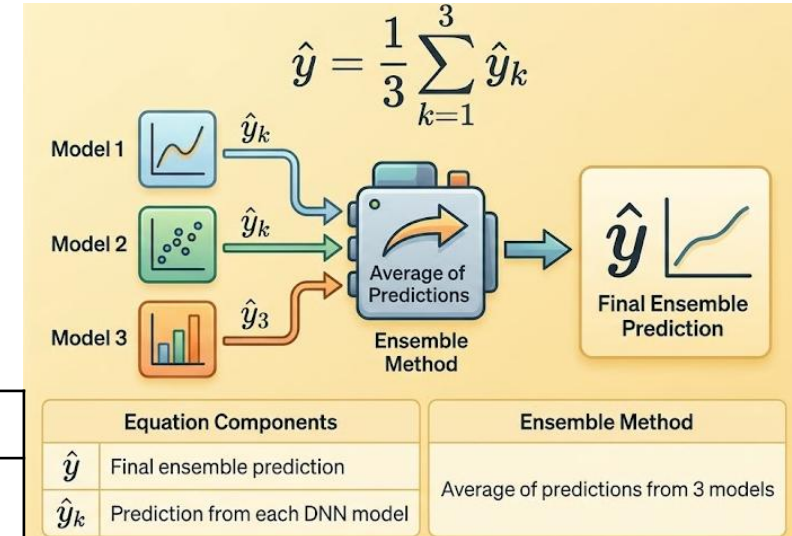
Purpose

- Prediction Tasks
- Tensile Strength Prediction
- Compressive Strength Prediction

Models Used

Model	Architecture	Function
Baseline DNN	Input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ Output	Standard nonlinear regression model
DNNWide	Wider hidden layers	Captures broad feature interactions
DNNDeep	5 hidden layers	Learns complex nonlinear relationships

Ensemble Learning



Ensemble learning improves stability, reduces overfitting, and increases prediction accuracy by combining multiple models for better generalized performance.



Branch B: Multi-Task Structural Integrity Prediction

1. Performs simultaneous structural property prediction and integrity assessment of CFRP composites
2. Uses a **Multi-Task Deep Neural Network (MT-DNN)** architecture
3. Shared backbone with multiple task-specific output heads
4. Learns interconnected structural properties together

Tasks Performed Simultaneously

1. Tensile Strength Regression
2. Fracture Toughness Regression
3. Structural Integrity Classification

Structural Integrity Classes

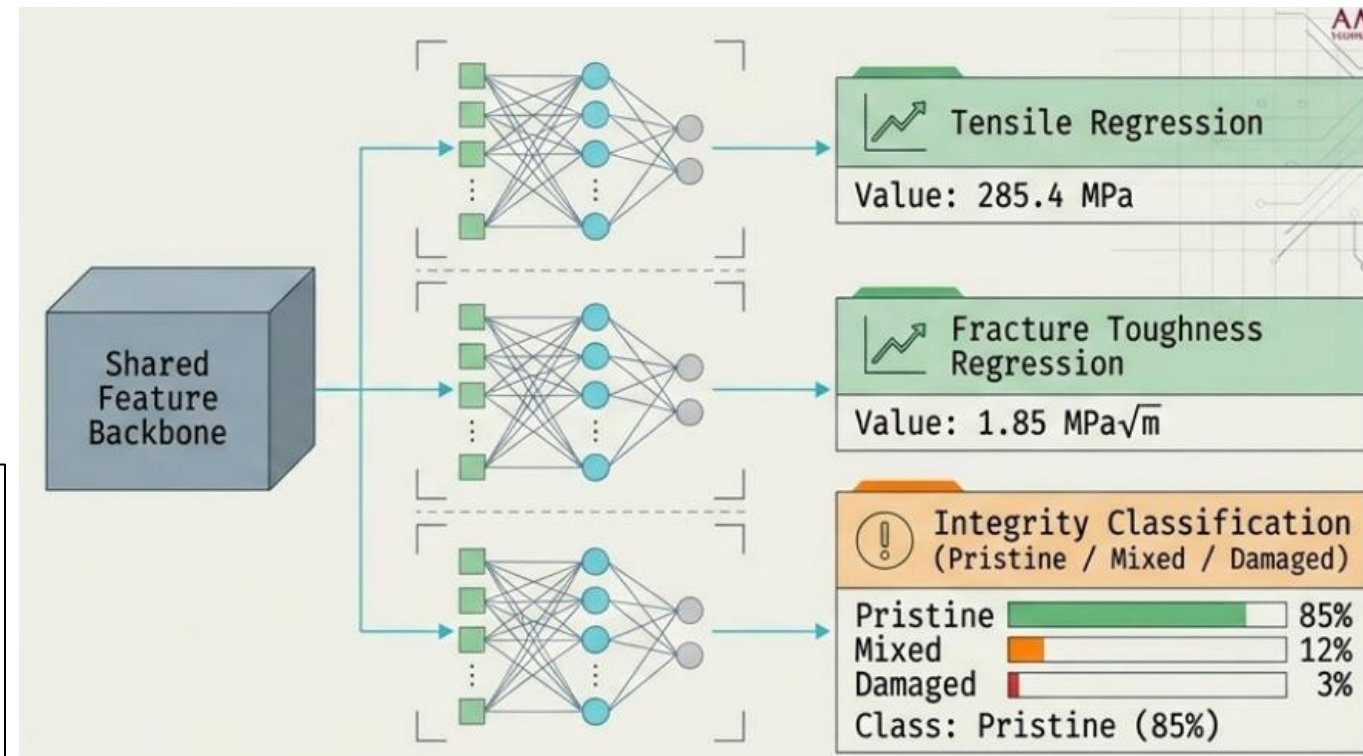
1. Pristine
2. Damaged
3. Mixed

Multi-Task Learning uses shared feature extraction to learn interconnected properties simultaneously, improving feature reuse, efficiency, robustness, and generalization while reducing computational complexity.

Input → Dense(256) → Dense(256) → Dense(128) → Dense(64)

Techniques Used

- a. GELU Activation
- b. Batch Normalization
- c. Dropout Regularization



Branch C: Hybrid Attention U-Net for Damage Segmentation

It performs pixel-wise damage segmentation on CFRP composite images using a Hybrid Attention U-Net architecture.

- crack detection
- delamination localization
- impact damage segmentation
- damaged region extraction

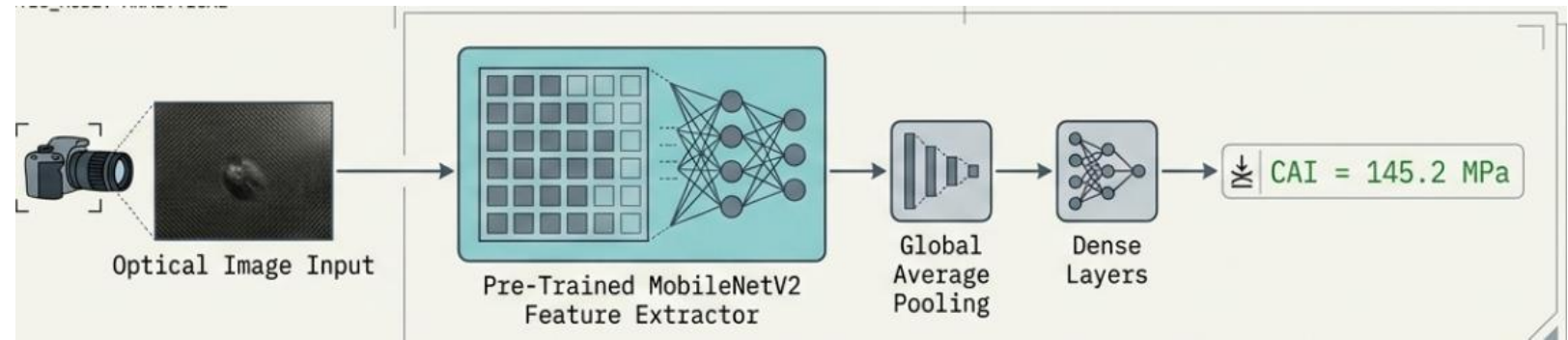
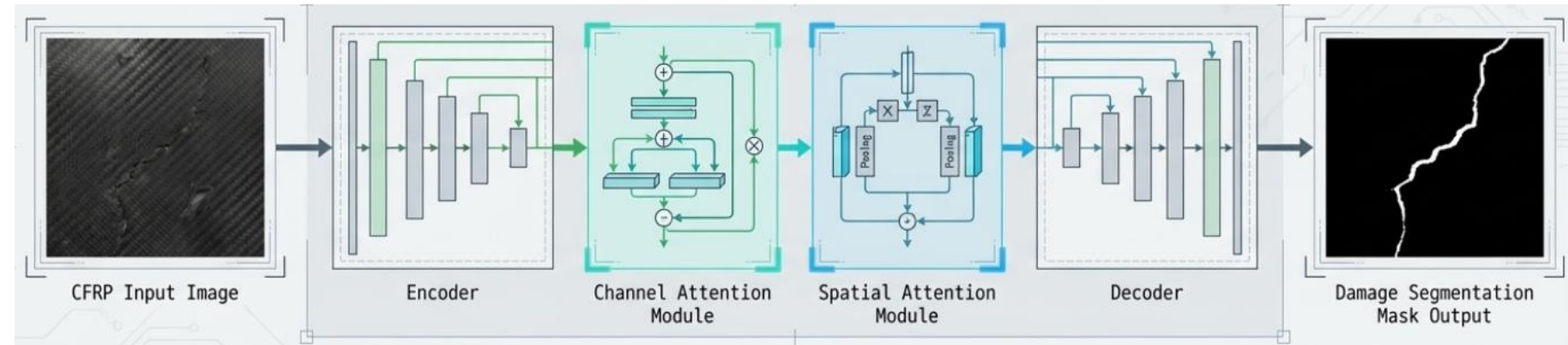
Image Preprocessing Pipeline Used

- Image resizing to 224×224
- Normalization to $[0,1]$
- CLAHE enhancement
- Denoising
- ROI extraction
- Otsu Threshold Masking

Role of Otsu Threshold Masking

- improve crack visibility
- isolate defect regions
- enhance damaged boundaries
- improve segmentation precision

The generated binary mask assists the segmentation model in learning clearer structural damage patterns.



Section	Content
Encoder Structure	The encoder extracts hierarchical damage features from CFRP images using convolution and pooling operations.
Encoder Filters	$32 \rightarrow 64 \rightarrow 128 \rightarrow 256$
Encoder Captures	• crack textures • delamination regions • damaged boundaries • fibre discontinuities at multiple feature scales.
Channel Attention Module	The channel attention mechanism helps the model identify the most important feature channels related to structural defects.
Enhances	• crack-related features • delamination textures • edge information while suppressing irrelevant background information.
Channel Attention Equation	$M_{ch} = \sigma \left(W_1 \delta \left(W_0 GAP(F) \right) + W_1 \delta \left(W_0 GMP(F) \right) \right)$
Spatial Attention Module	The spatial attention mechanism helps the network identify the exact damaged locations within the CFRP image.
Improves	• damage localization • boundary refinement • pixel-level segmentation accuracy
Spatial Attention Equation	$M_{sp} = \sigma \left(f_{7 \times 7} \left([AvgPool(F); MaxPool(F)] \right) \right)$
Final Attention Feature Refinement	The outputs of channel attention and spatial attention are combined to generate enhanced feature maps for accurate segmentation.
Final Attention Output	$F' = M_{sp} \otimes (M_{ch} \otimes F)$
This Improves	• segmentation precision • crack boundary detection • defect localization capability
Decoder Structure	The decoder reconstructs the segmentation mask using: • upsampling layers • skip connections • refined attention features
Final Output Produced	• binary damage segmentation mask representing damaged and non-damaged CFRP regions.



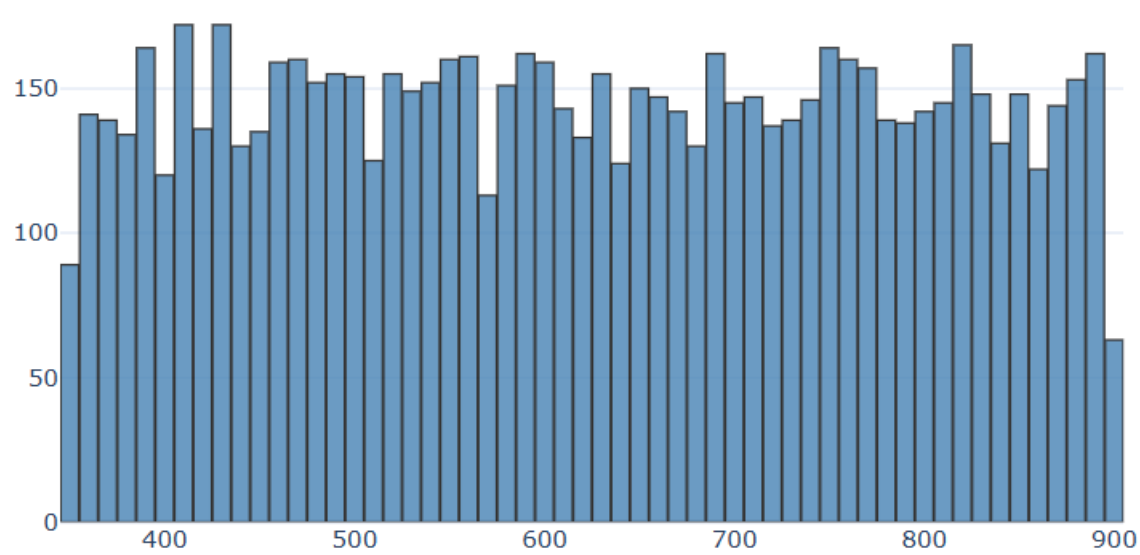
Training Strategy, Explainability & Final Workflow Training & Optimization

Training Configuration	Parameter		Value
	Optimizer		Adam
	Batch Size		8–256
	Epochs		20–100
	Learning Rate		1e–3 / 1e–5
	Regularization		Dropout + Weight Decay
Optimization Techniques	• Early stopping • ReduceLROnPlateau • Batch normalization • Fine-tuning • Two-phase training		
Explainable AI Integration	SHAP	Global feature importance	Identifies dominant material parameters
	LIME	Local prediction explanation	Instance-level interpretability
	Grad-CAM	Visual damage localization	Highlights crack regions in CFRP images
Final Engineering Outputs	Regression Outputs	• Tensile strength • Compressive strength • Fracture toughness • CAI strength	
	Classification & Segmentation Outputs	• Structural integrity class • Pixel-wise damage masks	

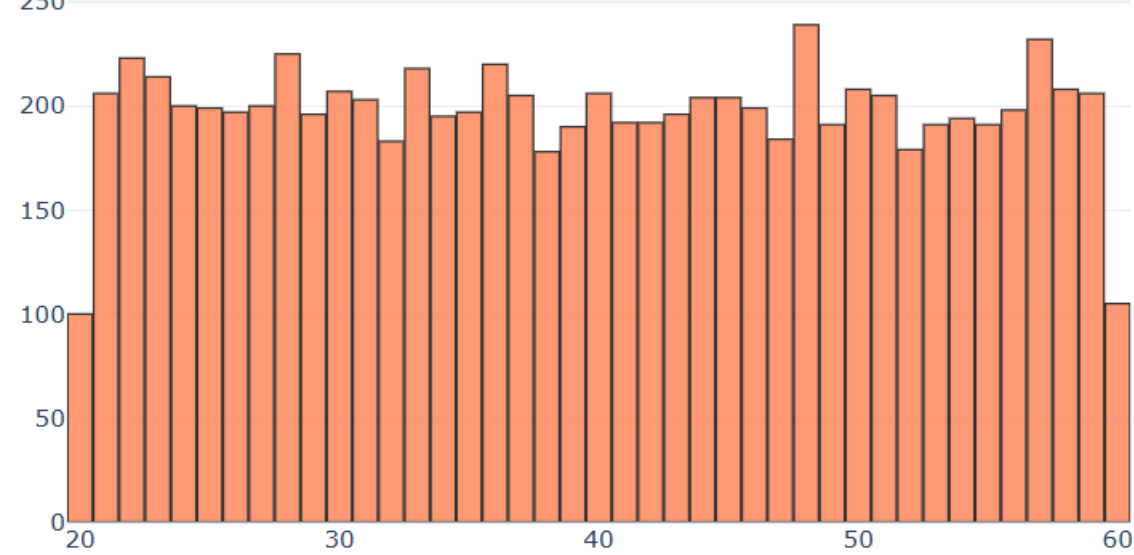
Category	Tools / Techniques Used
Programming & Frameworks	Python, PyTorch 2.x, TensorFlow/Keras 2.x, Google Colab
Datasets	Tabular Material Dataset, CFRP Damage Images, CAI Image Dataset
Preprocessing	IQR Outlier Removal, Normalization, Scaling, ROI Extraction, CLAHE, Denoising, Otsu Masking
Feature Engineering	Physics-Informed Compressive Strength Formulation
Branch A Models	DNN, DNNWide, DNNDeep, Ensemble Learning
Branch B Model	Multi-Task DNN
Branch C Model	Hybrid Attention U-Net
Branch D Model	MobileNetV2 Transfer Learning
Attention Mechanisms	Channel Attention, Spatial Attention
Activation Functions	ReLU, GELU, Sigmoid, Softmax
Optimization	Adam Optimizer, Early Stopping, ReduceLROnPlateau, Fine-Tuning
Regularization	Dropout, Batch Normalization, Weight Decay (L2)
Loss Functions	MSE, Binary Cross-Entropy, Cross-Entropy, Huber Loss
Explainable AI	SHAP, LIME, Grad-CAM
Evaluation Metrics	MAE, RMSE, R ² , Accuracy, Precision, Recall, F1-Score, Dice, IoU
Learning Approaches	Multimodal Learning, Multi-Task Learning, Transfer Learning, Explainable AI
Application Domain	CFRP Structural Health Monitoring & Damage Assessment

Target Variable Distributions

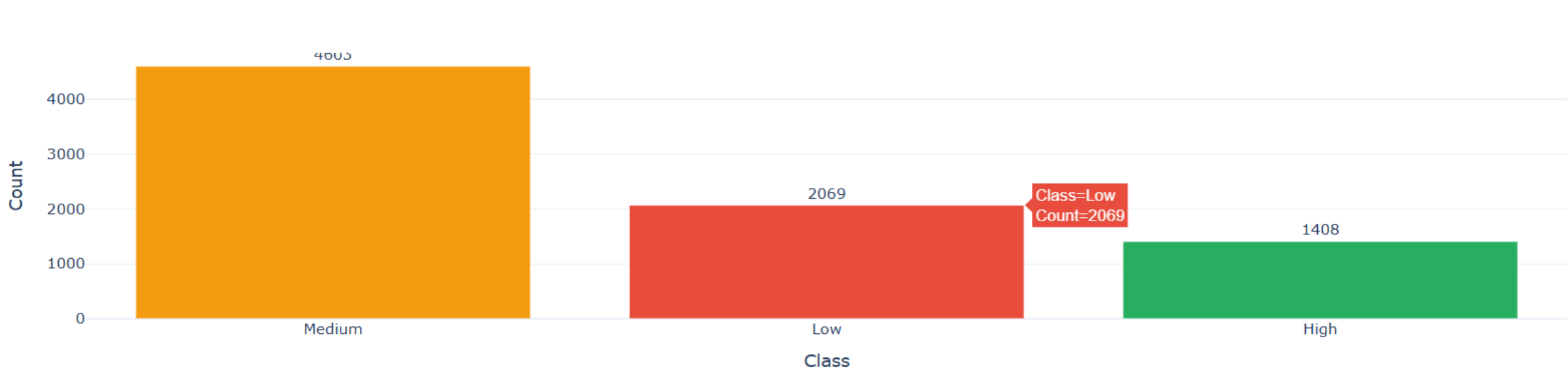
Tensile Strength Distribution (MPa)



Fracture Toughness Distribution (MPa√m)

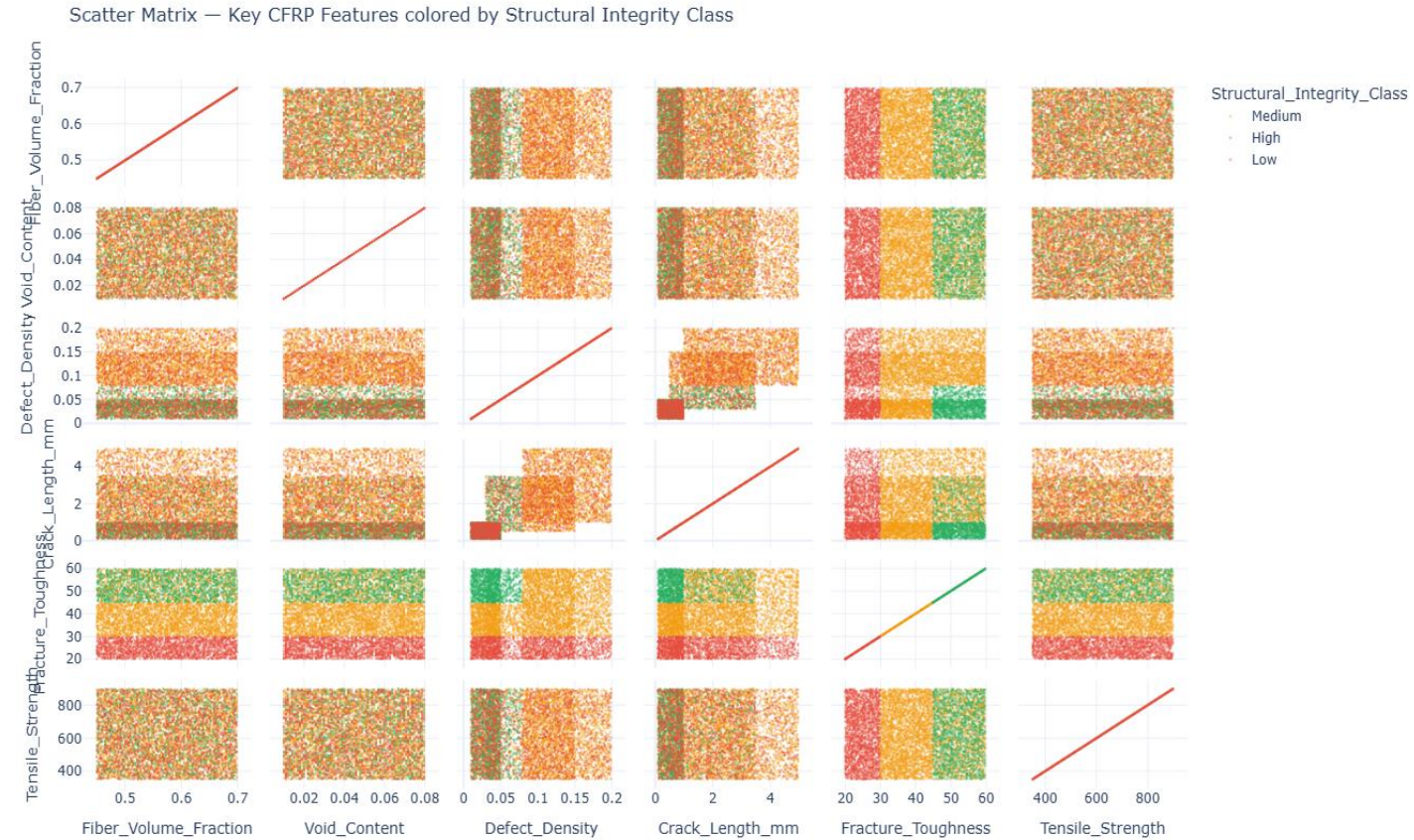
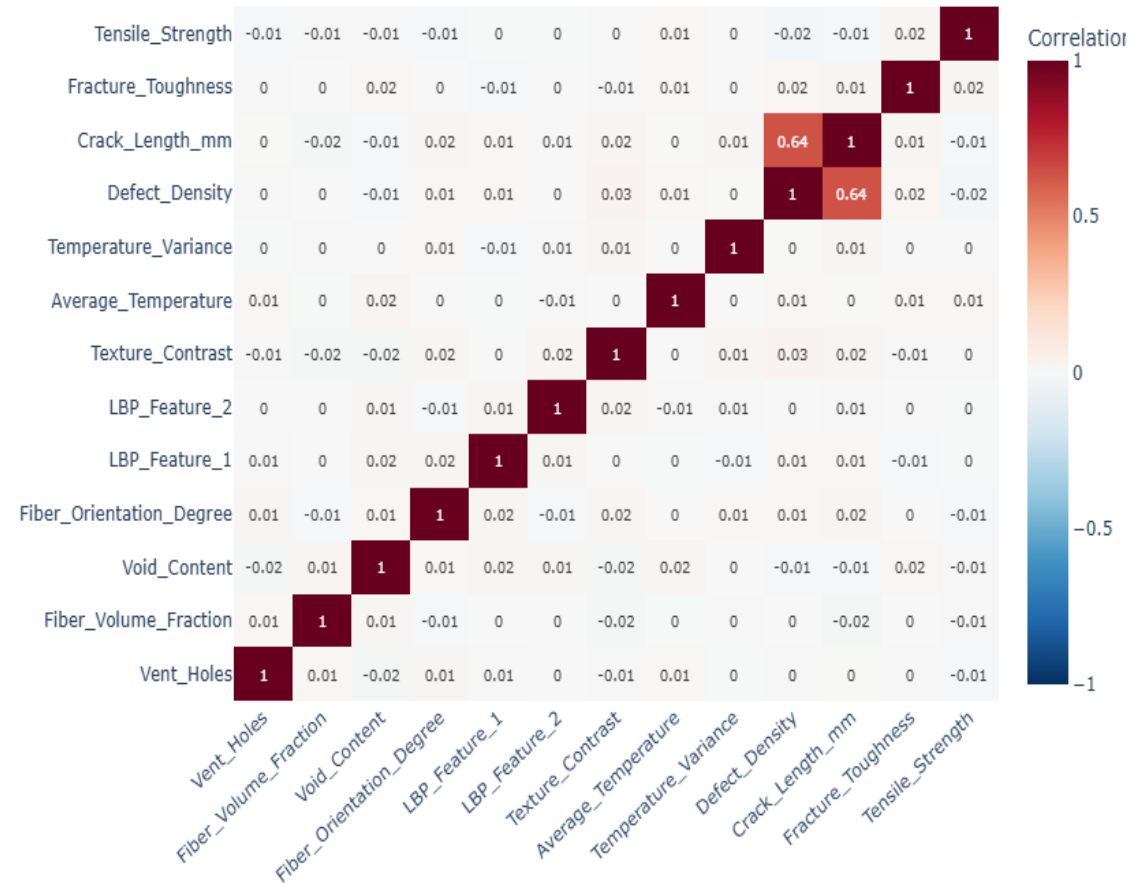


Structural Integrity Class Distribution



Results

Interactive Correlation Heatmap — CFRP Features





Results Branch A

Model 2:

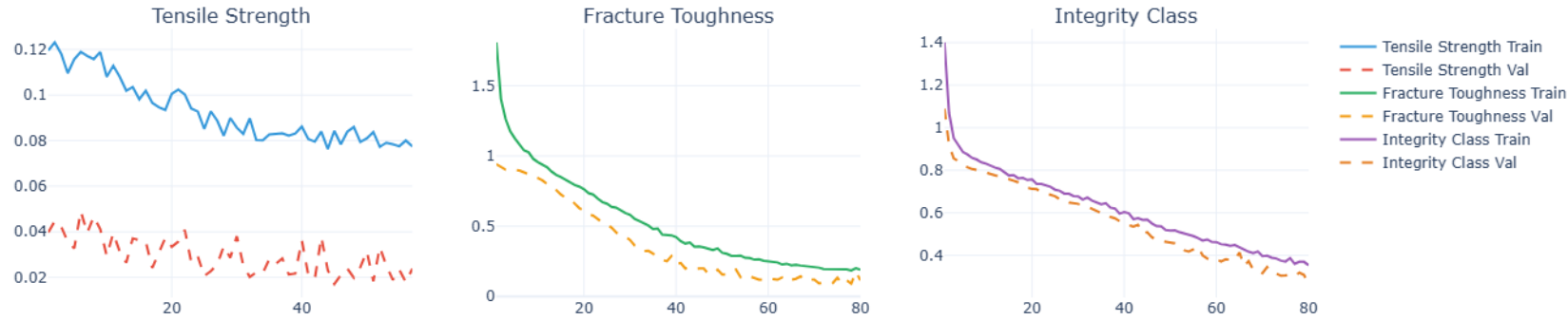
Model 1:

Epoch	Train Loss	Validation Loss	Learning Rate	Training Status / Observation
10	0.10804	0.02947	0.001000	Stable training progression
20	0.10051	0.03337	0.001000	Validation fluctuation observed
—	—	—	0.001000 → 0.000500	Learning rate reduced
30	0.08572	0.03805	0.000500	Improved optimization stability
—	—	—	0.000500 → 0.000250	Learning rate reduced
—	—	—	0.000250 → 0.000125	Learning rate reduced
40	0.08621	0.03628	0.000125	Better convergence achieved
50	0.08101	0.03140	0.000125	Reduced regression error
—	—	—	0.000125 → 0.000063	Learning rate reduced
—	—	—	0.000063 → 0.000031	Final learning rate reduction
57	—	0.01710	0.000031	Early stopping applied with best validation loss achieved

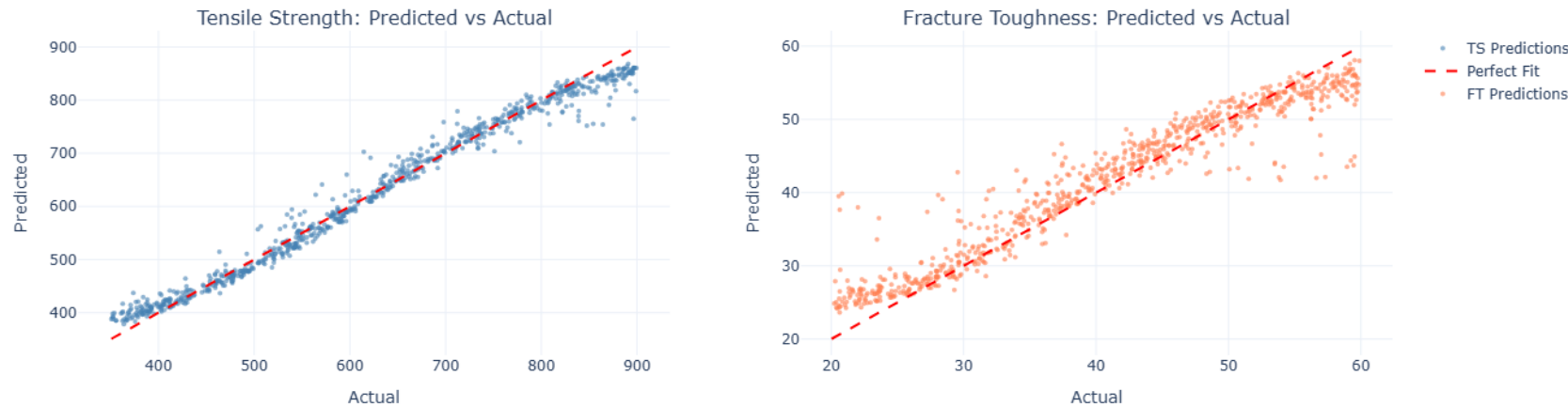
Epoch	Train Loss	Validation Loss	Learning Rate	Training Status / Observation
10	0.95651	0.84284	0.001000	Initial model convergence started
20	0.76389	0.61067	0.001000	Significant loss reduction observed
30	0.58161	0.40268	0.001000	Improved regression learning
40	0.42340	0.23106	0.001000	Better feature representation achieved
50	0.31148	0.15721	0.001000	Strong validation improvement
60	0.25167	0.13045	0.001000	Stable optimization observed
—	—	—	0.001000 → 0.000500	Learning rate reduced
70	0.20704	0.11902	0.000500	Improved convergence stability
—	—	—	0.000500 → 0.000250	Learning rate reduced
80	0.19019	0.11590	0.000250	Final optimized regression performance
Final	—	0.08906	0.000250	Best validation loss achieved

Results Branch A

Training & Validation Loss Curves



Model Predictions vs Ground Truth | TS $R^2=0.980$ | FT $R^2=0.904$



TENSILE STRENGTH (Regression)

R^2 Score : 0.9804
 RMSE : 22.227 MPa
 MAE : 16.008 MPa

FRACTURE TOUGHNESS (Regression)

R^2 Score : 0.9037
 RMSE : 3.578 MPa $\sqrt{m}$
 MAE : 2.494 MPa $\sqrt{m}$

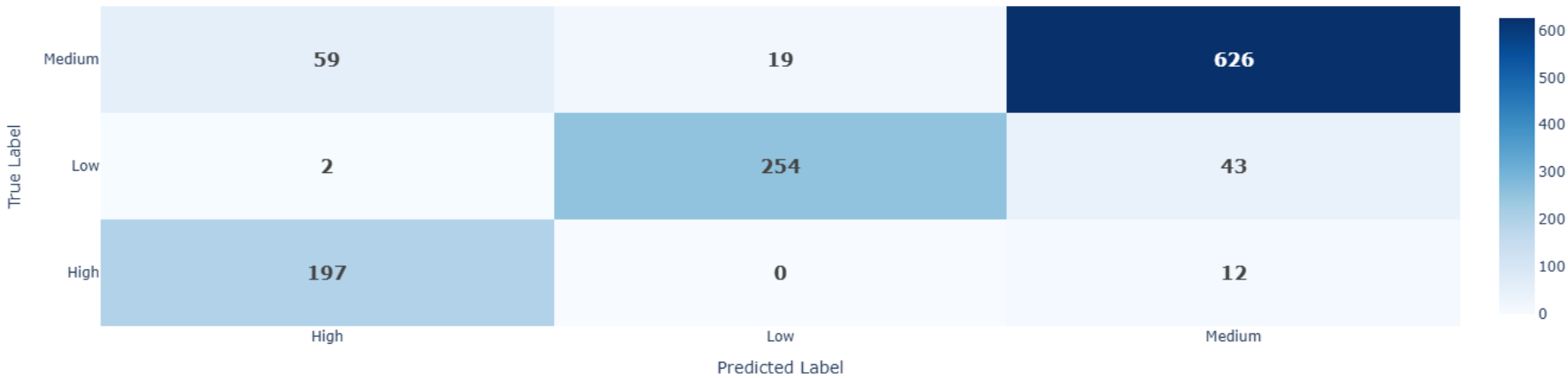
STRUCTURAL INTEGRITY CLASS (Classification)

	precision	recall	f1-score	support
High	0.76	0.94	0.84	209
Low	0.93	0.85	0.89	299
Medium	0.92	0.89	0.90	704
accuracy			0.89	1212
macro avg	0.87	0.89	0.88	1212
weighted avg	0.90	0.89	0.89	1212

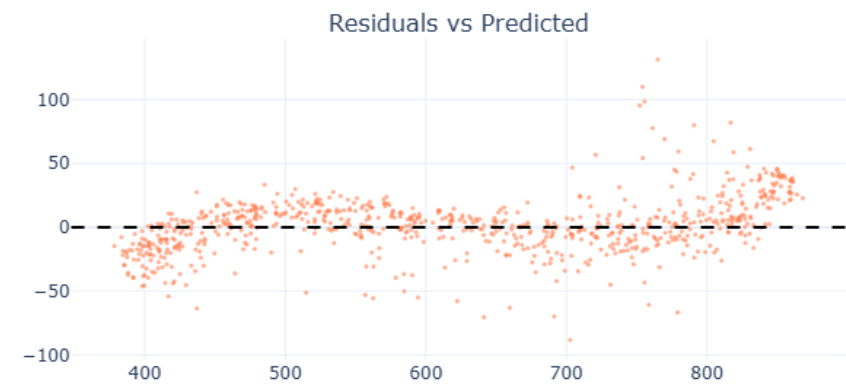
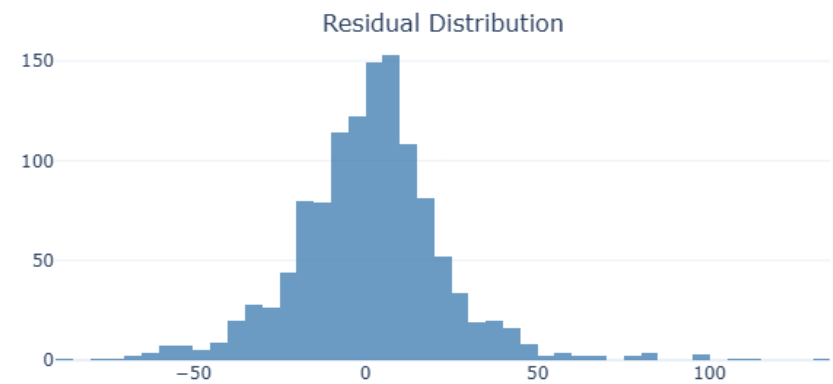


Results Branch A

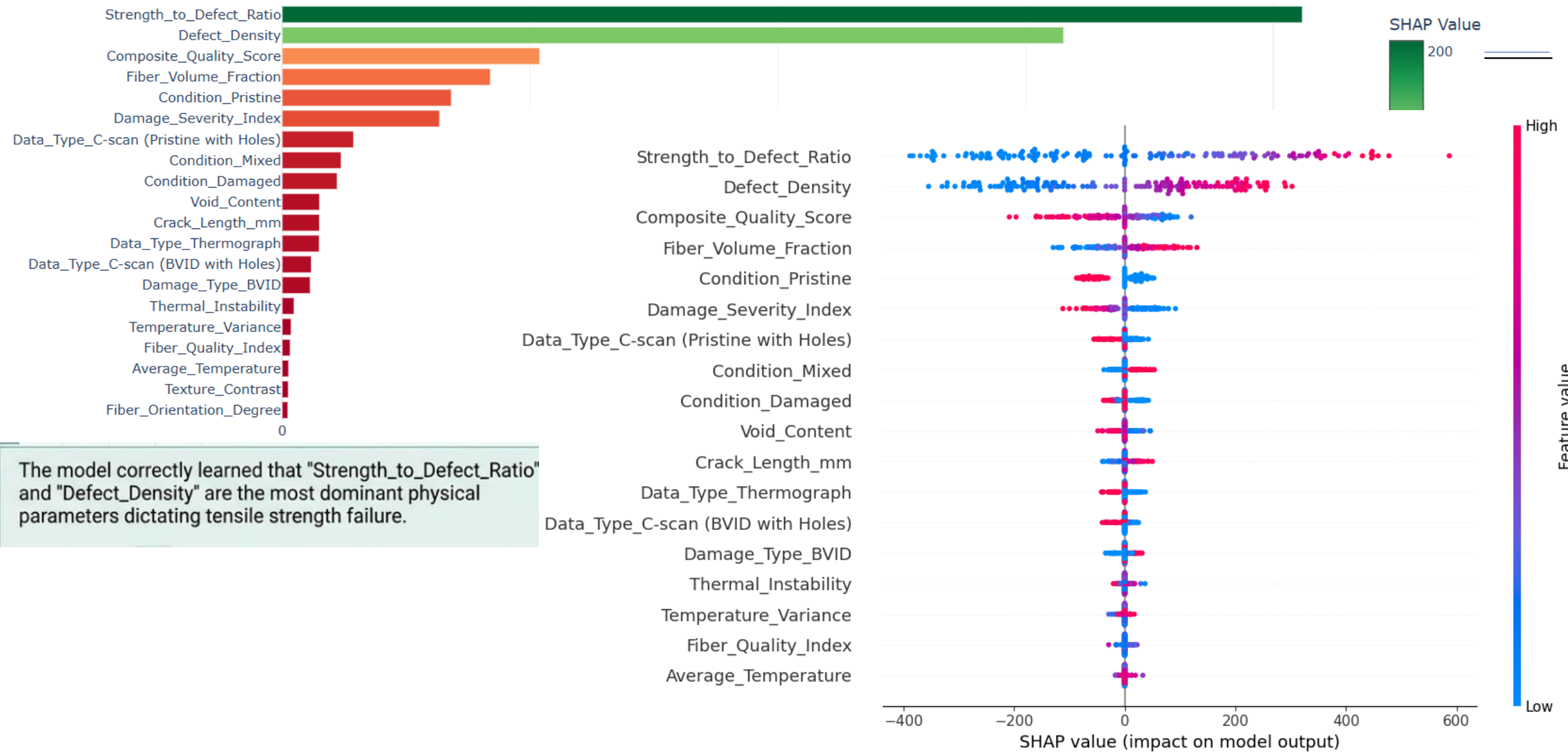
Confusion Matrix — Structural Integrity Classification | Acc=0.889



Residual Analysis — Tensile Strength Model



SHAP Feature Importance — Tensile Strength Model
Higher value = stronger influence on predictions

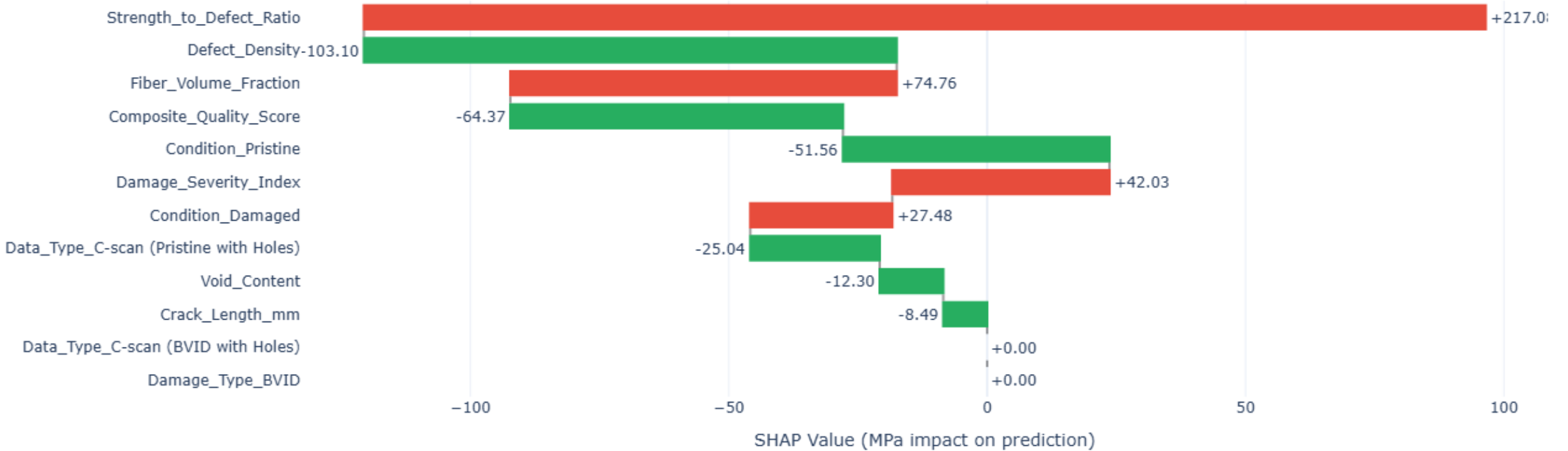




Results Branch A

SHAP Waterfall — Single Sample Explanation

Predicted Tensile Strength: 722.5 MPa | Baseline: 626.0 MPa | Red = increases prediction, Green = decreases





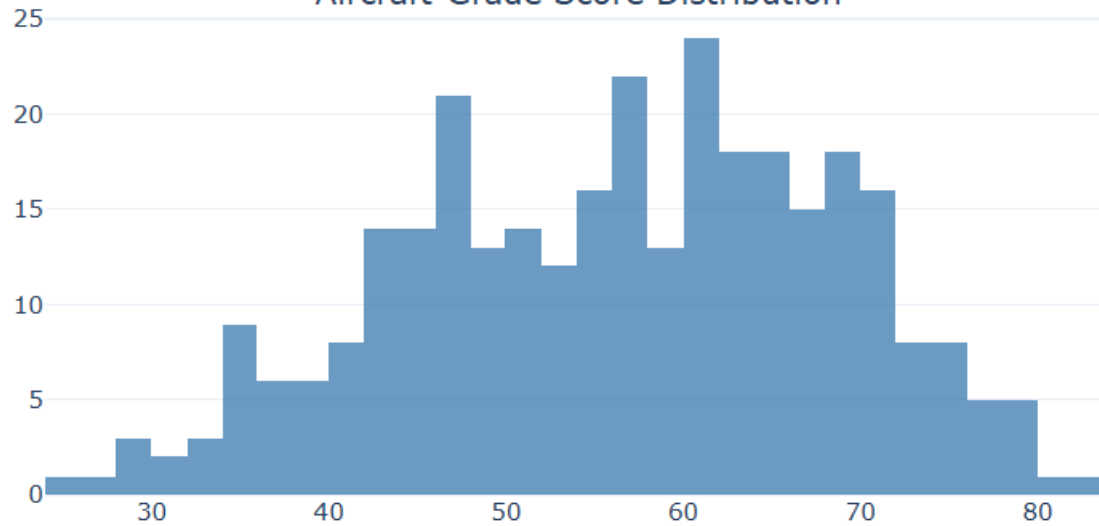
Results Branch A

Predictions generated for all samples						
	Tensile_Strength	Predicted_Tensile_Strength	Fracture_Toughness	Predicted_Fracture_Toughness	Structural_Integrity_Class	Predicted_Integrity_Class
0	372.62	397.718231	31.30	30.944988	Medium	Medium
1	413.52	418.784058	49.43	49.569141	Medium	Medium
2	497.68	495.871002	34.65	34.747303	Medium	Medium
3	505.09	478.651001	31.30	30.562855	Medium	Medium
4	515.48	510.959167	59.72	45.943970	High	High
5	624.25	686.616211	39.62	39.719067	Medium	Medium
6	476.49	476.394043	37.17	37.833710	Medium	Medium
7	527.65	508.368988	33.81	33.917038	Medium	Medium
8	540.22	521.326599	23.46	33.586563	Low	Low
9	470.54	466.558136	35.87	34.294540	Medium	Medium

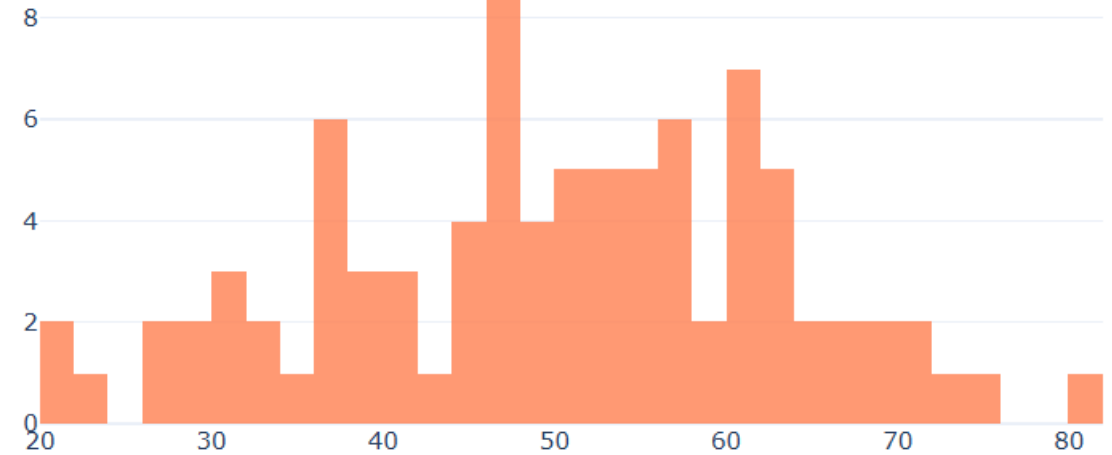
Total CFRP samples : 8,080 Aircraft-grade candidates : 315 (3.9%) Spacecraft-grade candidates: 89 (1.1%)	Model	Metric 1	Metric 2	Metric 3	Performance
	Tensile Strength Regression	$R^2 = 0.9804$	RMSE = 22.227 MPa	MAE = 16.008 MPa	Excellent
	Fracture Toughness Regression	$R^2 = 0.9037$	RMSE = 3.578 MPa√m	MAE = 2.494 MPa√m	Very Good
	Structural Integrity Classification	Accuracy = 0.89	Macro F1 = 0.88	Weighted F1 = 0.89	High Accuracy

Aerospace Score Distribution of Qualifying CFRP Samples

Aircraft-Grade Score Distribution



Spacecraft-Grade Score Distribution



Optimal Composition for Spacecraft CFRP:

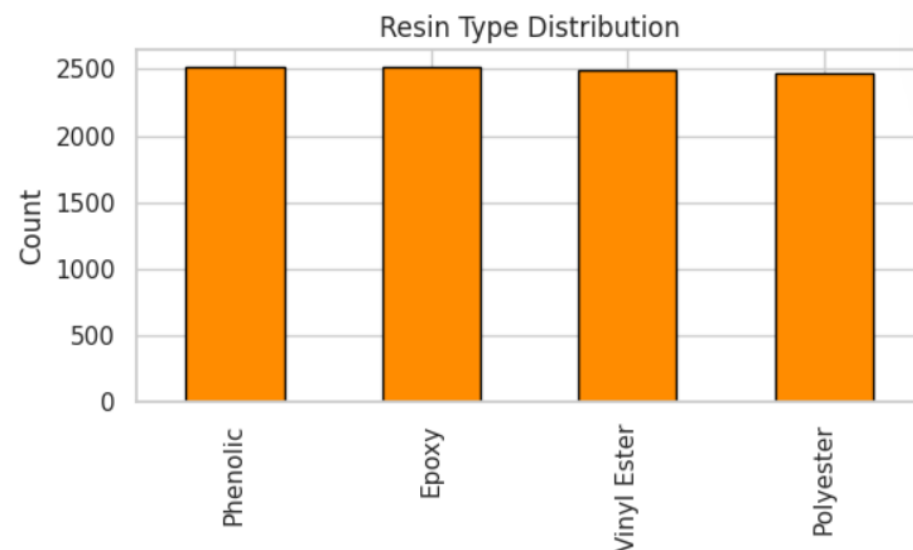
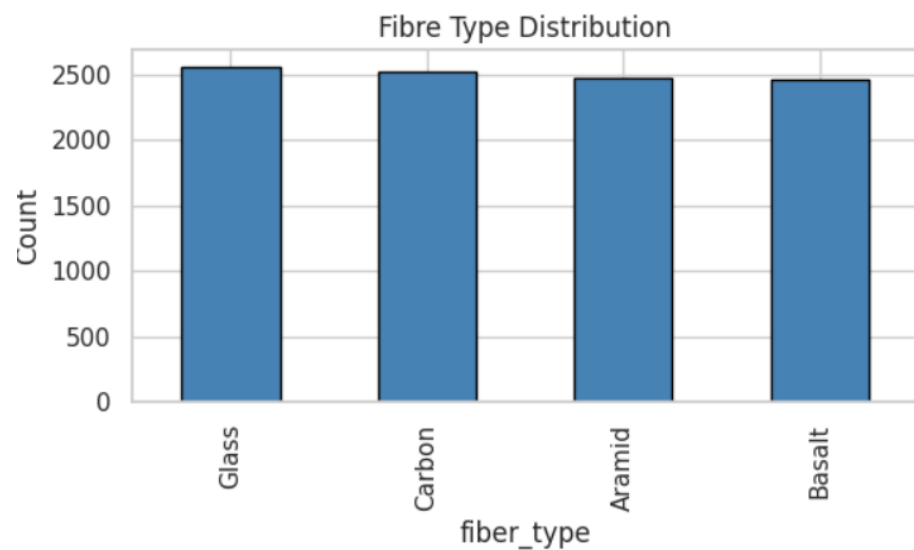
Fiber Volume Fraction : 0.510
 Void Content : 0.0390
 Predicted TS : 82.5 MPa

EXPLAINING TOP SPACECRAFT-GRADE CFRP SAMPLE

Sample ID : 5571
 Condition : Pristine
 Fiber Vol. Frac. : 0.584
 Void Content : 0.0140
 Defect Density : 0.0400
 Crack Length (mm) : 0.961
 Fiber Orientation : 25.4°
 Avg Temperature : 108.9°C
 Actual TS : 890.3 MPa
 Predicted TS : 861.3 MPa
 Actual FT : 58.61 MPa√m
 Predicted FT : 56.21 MPa√m
 Integrity Class : High
 Aerospace Score : 80.45/100

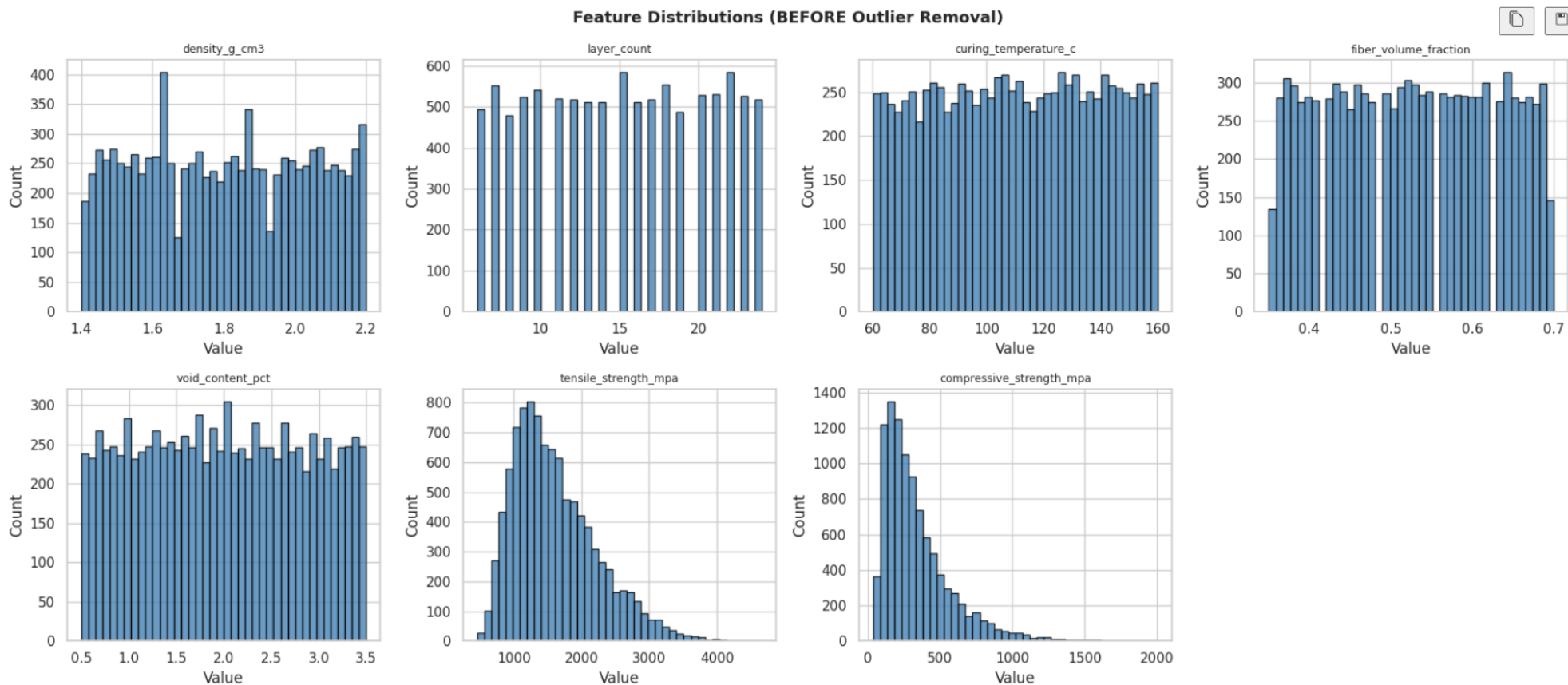
This sample qualifies for SPACECRAFT use because:

- ✓ Tensile Strength 861 MPa $\geq$ 700 MPa
- ✓ Fracture Toughness 56.2 MPa√m $\geq$ 45 MPa√m
- ✓ Defect Density 0.0400 $\leq$ 0.05
- ✓ Void Content 0.0140 $\leq$ 0.03
- ✓ Crack Length 0.961 mm $\leq$ 1.0 mm
- ✓ Structural Integrity: High

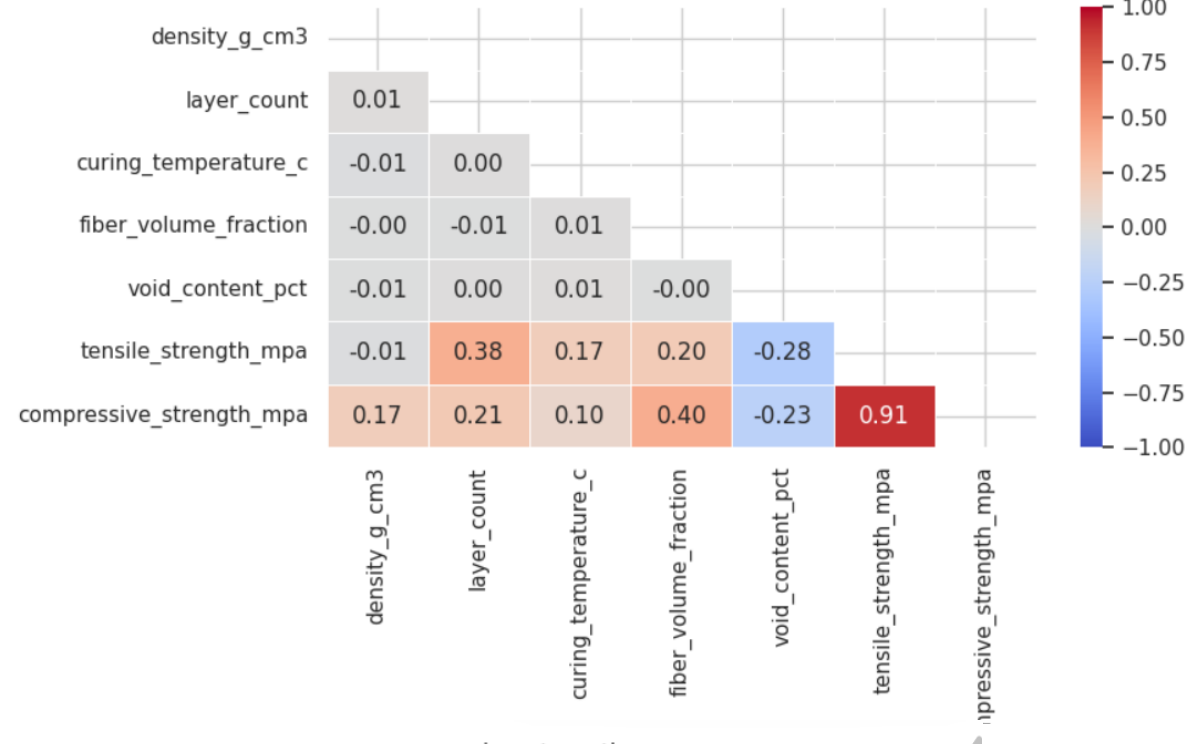


Compressive Strength stats:

count	10000.00
mean	335.07
std	234.42
min	36.03
25%	168.08
50%	269.58
75%	431.86
max	2006.87



Feature Correlation Matrix (Before Outlier Removal)



Rows BEFORE outlier removal: 10000

[tensile_strength_mpa] Q1=1135.0, Q3=1968.2, IQR=833.2 → fences=[-114.8, 3218.0] outliers=168

[compressive_strength_mpa] Q1=168.1, Q3=431.9, IQR=263.8 → fences=[-227.6, 827.5] outliers=454

Rows AFTER outlier removal: 9531

Rows removed: 469 (4.7%)

Fibre type encoding: {'Aramid': np.int64(0), 'Basalt': np.int64(1), 'Carbon': np.int64(2), 'Glass': np.int64(3)}

Resin type encoding: {'Epoxy': np.int64(0), 'Phenolic': np.int64(1), 'Polyester': np.int64(2), 'Vinyl Ester': np.int64(3)}

Feature matrix X: (9531, 7)

Target matrix y: (9531, 2)

Feature scaling done ✔

X mean after scaling: [0. -0. -0. -0. 0. -0. -0.]

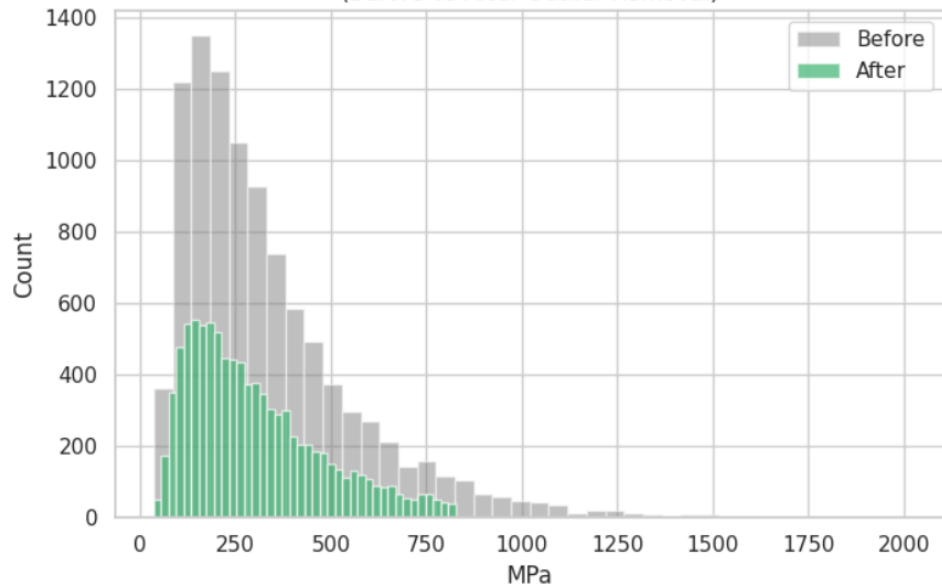
X std after scaling: [1. 1. 1. 1. 1. 1. 1.]

Train : 6675 samples

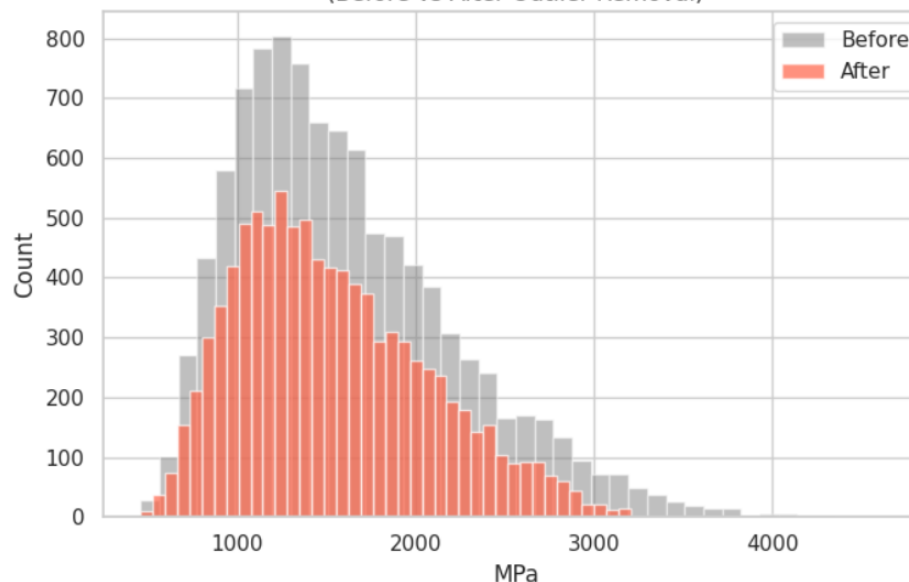
Val : 1426 samples

Test : 1430 samples

**compressive_strength_mpa
(Before vs After Outlier Removal)**



**tensile_strength_mpa
(Before vs After Outlier Removal)**

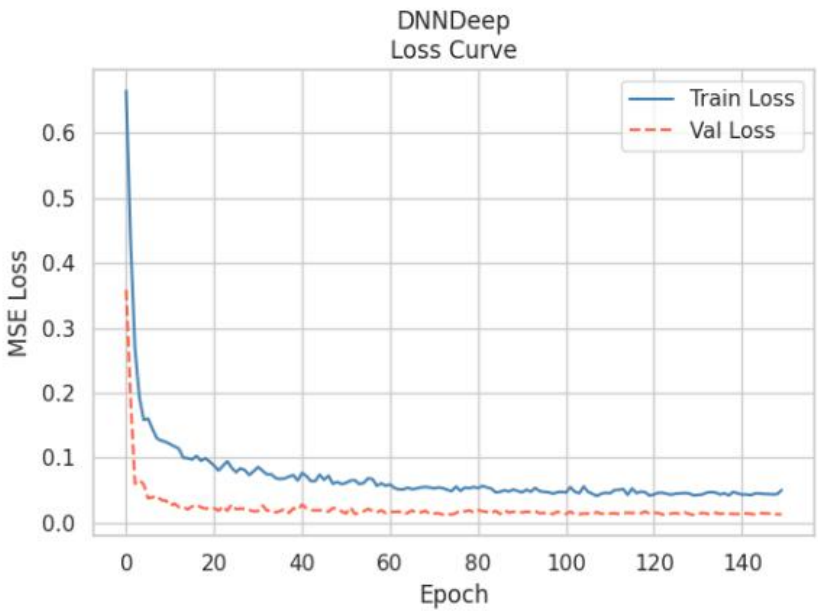
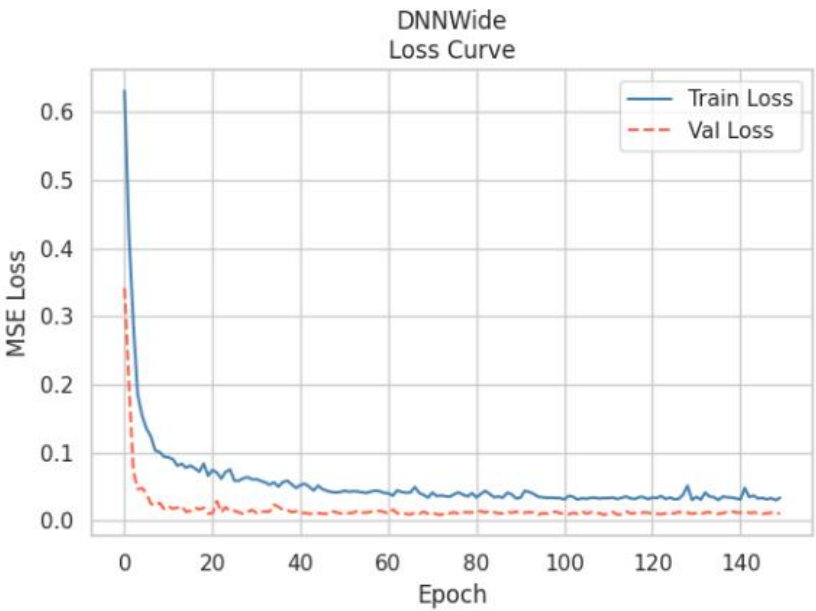
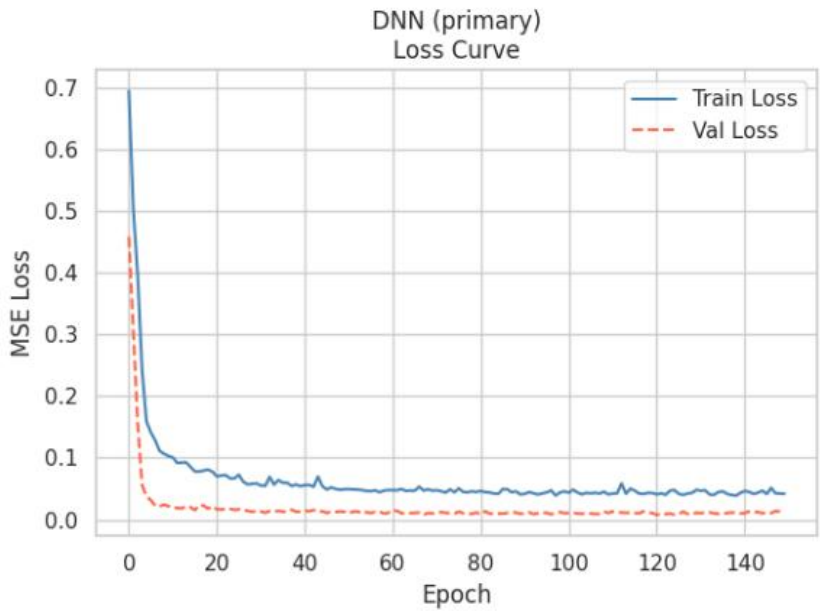




Results Branch A

Model	Final Epoch	Train Loss	Validation Loss	Performance
DNN (Primary)	150/150	0.0419	0.0109	Best Overall
DNNWide	150/150	0.0333	0.0107	Stable Ensemble Learning
DNNDeep	150/150	0.0499	0.0130	Deep Feature Learning

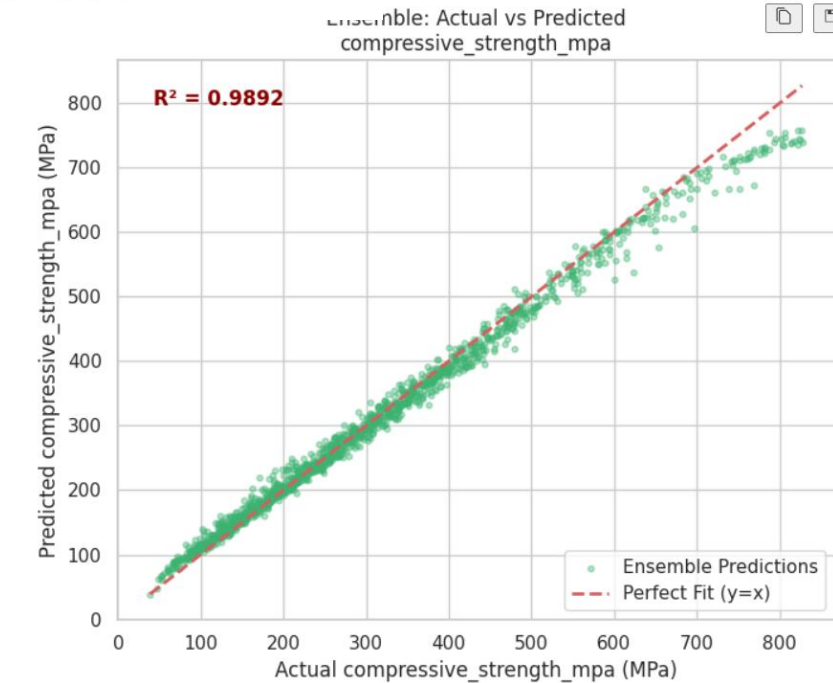
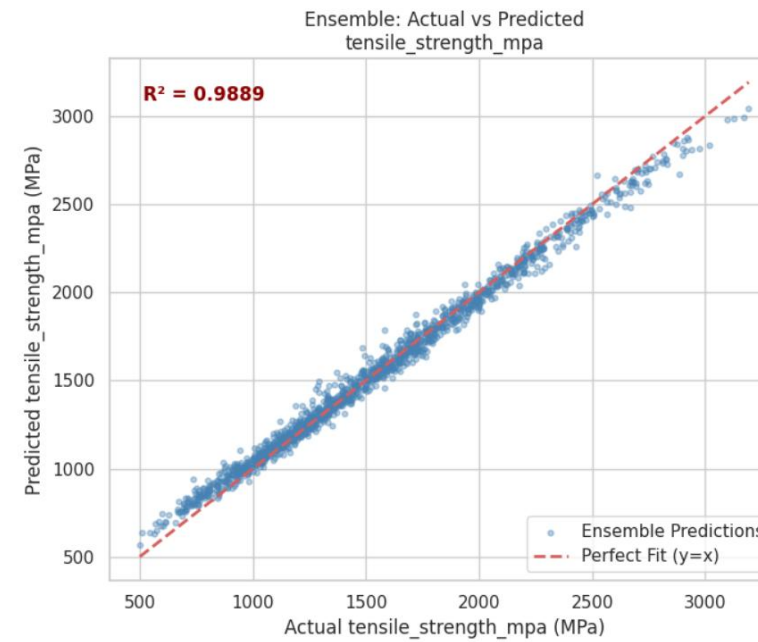
Training vs Validation Loss Curves





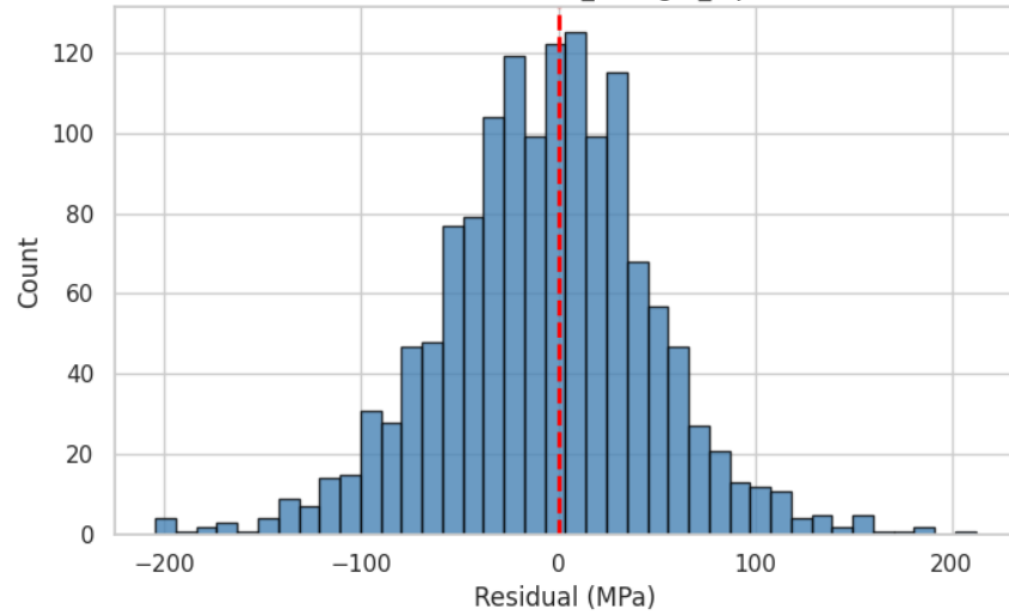
Results Branch A

Model	Target Property	MAE	RMSE	R ² Score
DNN (Primary)	Tensile Strength	41.13 MPa	53.42 MPa	0.9895
	Compressive Strength	13.69 MPa	19.75 MPa	0.9869
DNNWide	Tensile Strength	44.13 MPa	57.08 MPa	0.9880
	Compressive Strength	12.85 MPa	18.13 MPa	0.9890
DNNDeep	Tensile Strength	46.97 MPa	61.34 MPa	0.9862
	Compressive Strength	14.54 MPa	19.56 MPa	0.9872
Ensemble (DNN + Wide + Deep)	Tensile Strength	42.20 MPa	54.87 MPa	0.9889
	Compressive Strength	12.65 MPa	17.98 MPa	0.9892

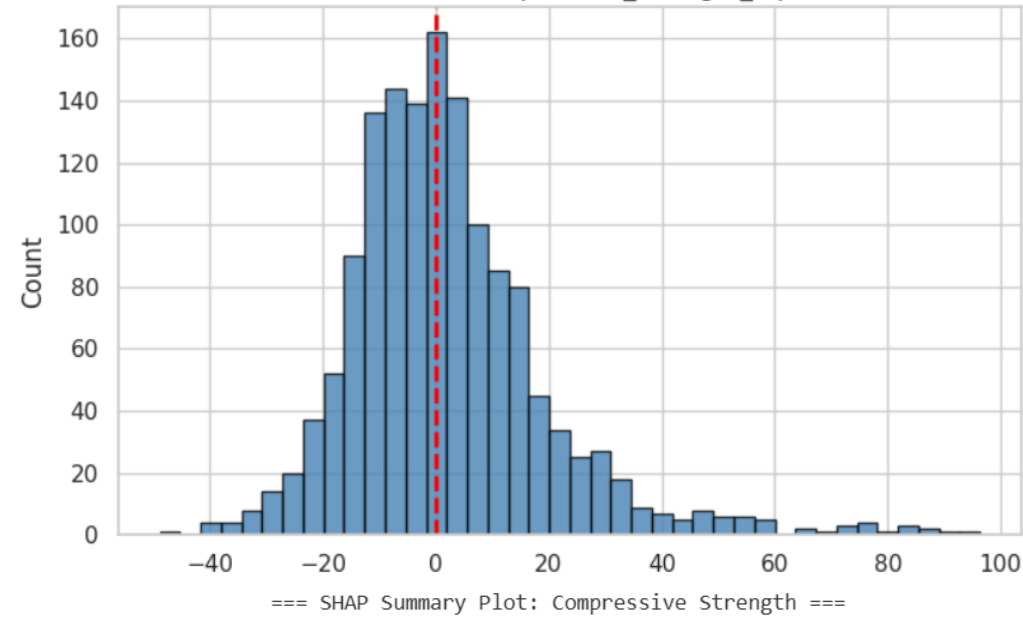


Ensemble Model — Residual Distributions

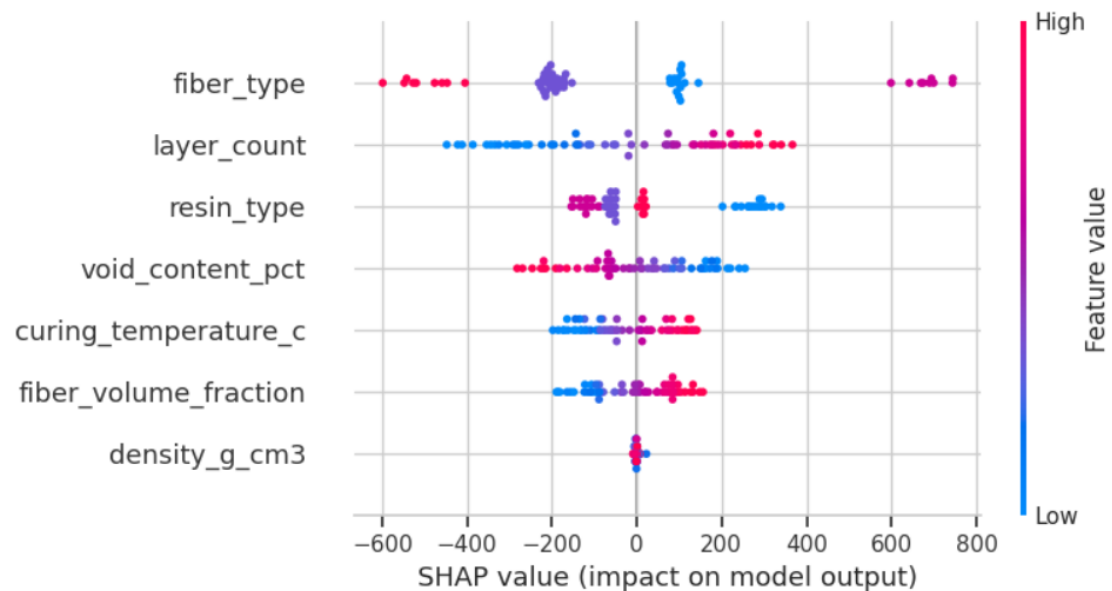
Residuals: tensile_strength_mpa



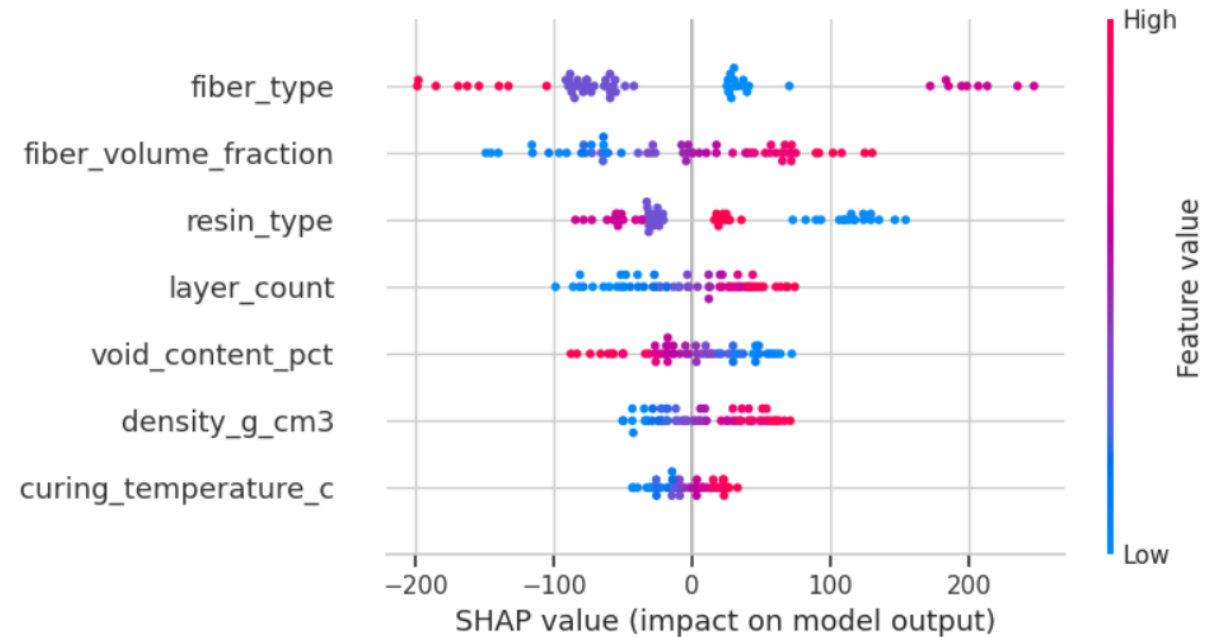
Residuals: compressive_strength_mpa



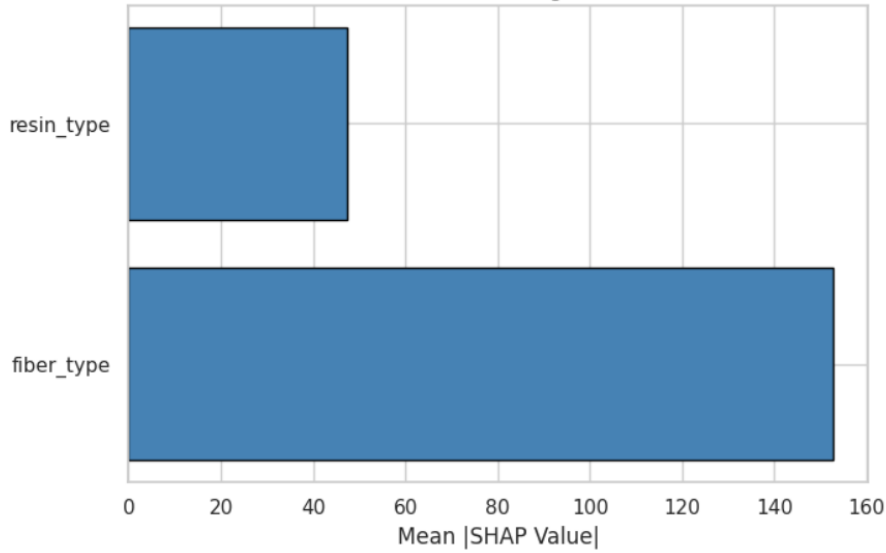
=== SHAP Summary Plot: Tensile Strength ===



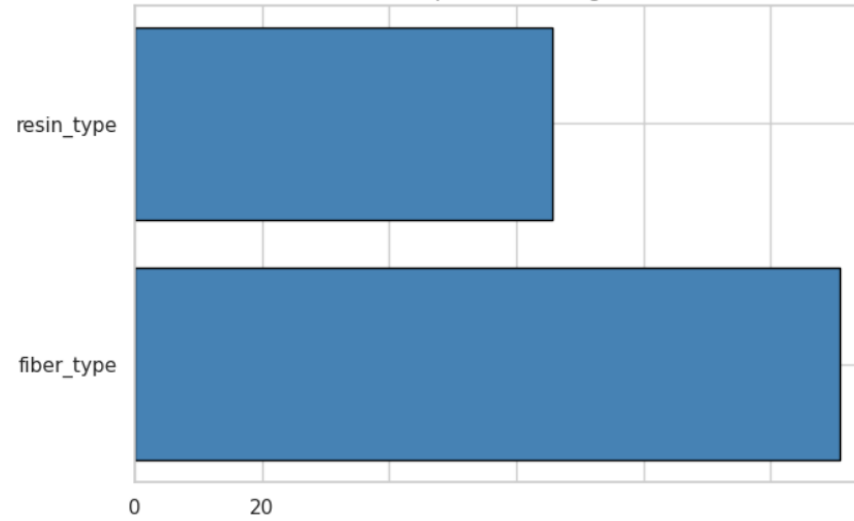
=== SHAP Summary Plot: Compressive Strength ===



SHAP Feature Importance
Tensile Strength



SHAP Feature Importance
Compressive Strength



LIME explanation for Test Sample 5 – Tensile Strength:

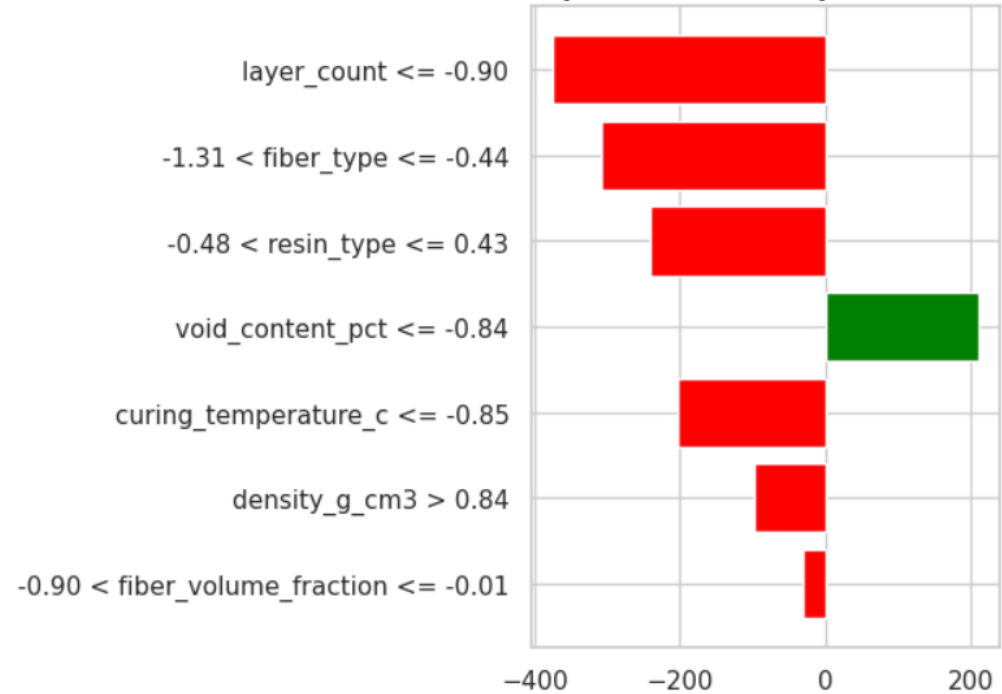
Model Prediction : 1011.14 MPa

Actual Value : 970.00 MPa

LIME Feature Contributions (positive = increases prediction):

layer_count <= -0.90	→ -376.6357
-1.31 < fiber_type <= -0.44	→ -309.0497
-0.48 < resin_type <= 0.43	→ -241.1996
void_content_pct <= -0.84	→ +211.0369
curing_temperature_c <= -0.85	→ -203.1959
density_g_cm3 > 0.84	→ -97.7140
-0.90 < fiber_volume_fraction <= -0.01	→ -29.7086

LIME Local Explanation — Sample 5 (Tensile Strength)



Total material candidates generated: 3,888

🏆 Top 15 Optimised Composite Material Combinations:

	fiber_type	resin_type	density_g_cm3	layer_count	curing_temperature_c	fiber_volume_fraction	void_content_pct	pred_tensile_mpa	pred_compressive_mpa	composite_score
0	Carbon	Epoxy	1.5	24	120	0.55	0.5	3289.100098	779.400024	0.9290
1	Carbon	Epoxy	1.5	16	160	0.55	0.5	3258.899902	790.000000	0.9261
2	Carbon	Epoxy	1.5	24	160	0.55	0.5	3303.500000	760.299988	0.9260
3	Carbon	Epoxy	1.5	24	160	0.55	1.5	3291.800049	759.500000	0.9233
4	Carbon	Epoxy	1.5	24	160	0.70	2.5	3145.800049	832.099976	0.9159
5	Carbon	Epoxy	1.5	24	120	0.70	1.5	3115.000000	849.700012	0.9150
6	Carbon	Epoxy	1.5	24	80	0.55	0.5	3218.000000	781.299988	0.9148
7	Carbon	Epoxy	1.5	24	160	0.70	1.5	3128.300049	837.000000	0.9138
8	Carbon	Epoxy	1.5	16	160	0.70	0.5	3088.300049	857.000000	0.9118
9	Carbon	Epoxy	1.5	16	160	0.70	1.5	3099.000000	846.900024	0.9108
10	Carbon	Epoxy	1.5	24	120	0.55	1.5	3219.300049	760.500000	0.9085
11	Carbon	Epoxy	1.5	24	160	0.40	0.5	3370.300049	658.599976	0.9078
12	Carbon	Vinyl Ester	1.5	24	160	0.70	0.5	3134.300049	813.299988	0.9075
13	Carbon	Epoxy	1.5	16	120	0.70	0.5	3059.199951	862.000000	0.9073
14	Carbon	Epoxy	1.5	24	80	0.70	0.5	3058.300049	858.500000	0.9060

Model	Tensile R ²	Compressive R ²	Notes
DNN (primary)	~0.987	~0.990	Fast, lightweight
DNNWide	~0.986	~0.989	Wider layers
DNNDeep	~0.986	~0.989	More depth
Ensemble	~0.989	~0.991	Best performer



Results Branch A

PRODUCT-SPECIFIC MATERIAL RECOMMENDATIONS

Material	σ_t (MPa)	σ_c (MPa)
Carbon/Epoxy Vf=0.65 n=20 T=150°C	3126.5	850.4
Glass/Polyester Vf=0.4 n=8 T=70°C	641.8	47.0
Aramid/Vinyl Ester Vf=0.55 n=12 T=110°C	1682.4	363.2
Basalt/Epoxy Vf=0.6 n=16 T=130°C	1996.9	502.2

♦ Helmet

Constraints: density ≤ 1.8 g/cm³, tensile ≥ 1200 MPa, compressive ≥ 200 MPa

- #1: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=120°C | $\sigma_t=3289.10009765625$ MPa | $\sigma_c=779.4000244140625$ MPa
#2: Carbon fibre + Epoxy resin | Vf=0.55 | layers=16 | cure=160°C | $\sigma_t=3258.89990234375$ MPa | $\sigma_c=790.0$ MPa
#3: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=160°C | $\sigma_t=3303.5$ MPa | $\sigma_c=760.2999877929688$ MPa

♦ Bike Frame

Constraints: density ≤ 2.0 g/cm³, tensile ≥ 1500 MPa, compressive ≥ 300 MPa

- #1: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=120°C | $\sigma_t=3289.10009765625$ MPa | $\sigma_c=779.4000244140625$ MPa
#2: Carbon fibre + Epoxy resin | Vf=0.55 | layers=16 | cure=160°C | $\sigma_t=3258.89990234375$ MPa | $\sigma_c=790.0$ MPa
#3: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=160°C | $\sigma_t=3303.5$ MPa | $\sigma_c=760.2999877929688$ MPa

♦ Phone Case

Constraints: density ≤ 1.6 g/cm³, tensile ≥ 800 MPa, compressive ≥ 150 MPa

- #1: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=120°C | $\sigma_t=3289.10009765625$ MPa | $\sigma_c=779.4000244140625$ MPa
#2: Carbon fibre + Epoxy resin | Vf=0.55 | layers=16 | cure=160°C | $\sigma_t=3258.89990234375$ MPa | $\sigma_c=790.0$ MPa
#3: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=160°C | $\sigma_t=3303.5$ MPa | $\sigma_c=760.2999877929688$ MPa

♦ Sports Equipment

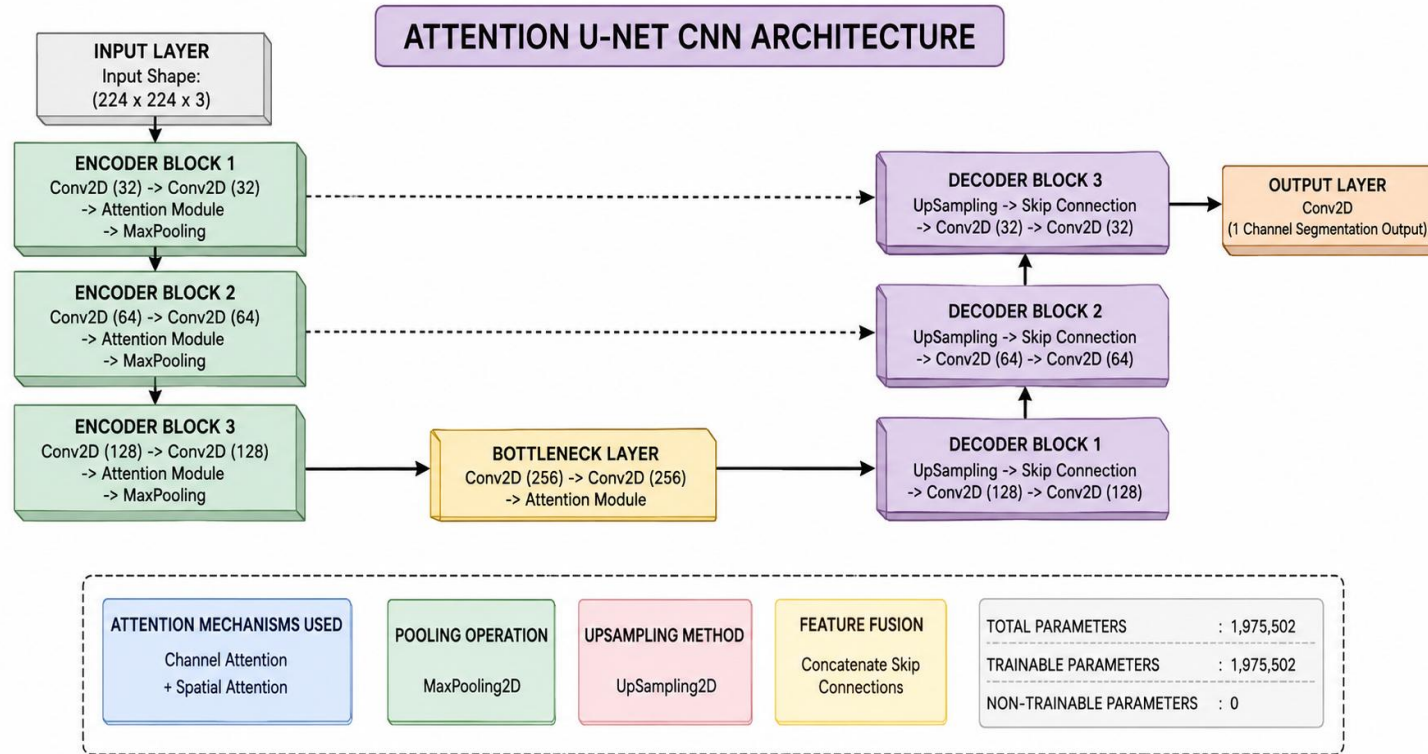
Constraints: density ≤ 2.1 g/cm³, tensile ≥ 1000 MPa, compressive ≥ 200 MPa

- #1: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=120°C | $\sigma_t=3289.10009765625$ MPa | $\sigma_c=779.4000244140625$ MPa
#2: Carbon fibre + Epoxy resin | Vf=0.55 | layers=16 | cure=160°C | $\sigma_t=3258.89990234375$ MPa | $\sigma_c=790.0$ MPa
#3: Carbon fibre + Epoxy resin | Vf=0.55 | layers=24 | cure=160°C | $\sigma_t=3303.5$ MPa | $\sigma_c=760.2999877929688$ MPa



Attention U-Net Based CNN Architecture for CFRP Damage Segmentation

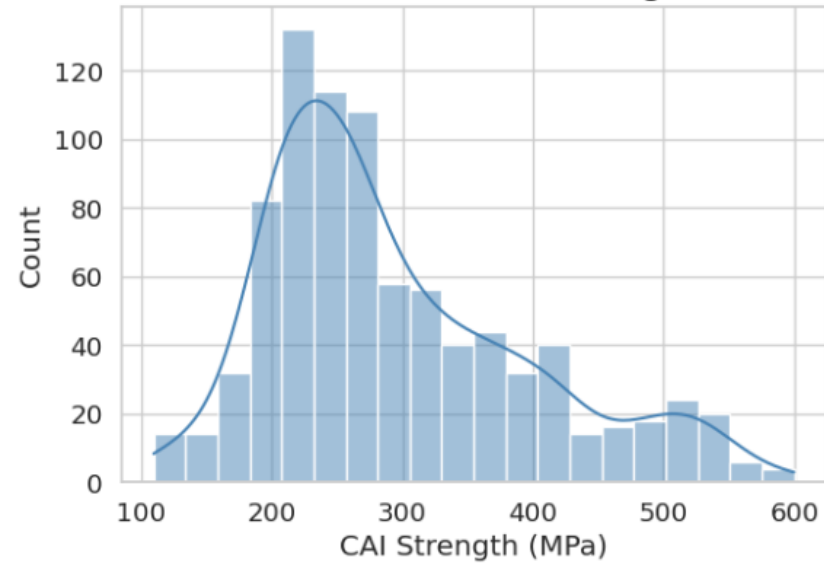
Section	Architecture Details
Model Name	Attention U-Net CNN Architecture
Input Layer	Input Shape: (224 x 224 x 3)
Encoder Block 1	Conv2D (32) -> Conv2D (32) -> Attention Module -> MaxPooling
Encoder Block 2	Conv2D (64) -> Conv2D (64) -> Attention Module -> MaxPooling
Encoder Block 3	Conv2D (128) -> Conv2D (128) -> Attention Module -> MaxPooling
Bottleneck Layer	Conv2D (256) -> Conv2D (256) -> Attention Module
Decoder Block 1	UpSampling -> Skip Connection -> Conv2D (128) -> Conv2D (128)
Decoder Block 2	UpSampling -> Skip Connection -> Conv2D (64) -> Conv2D (64)
Decoder Block 3	UpSampling -> Skip Connection -> Conv2D (32) -> Conv2D (32)
Attention Mechanisms Used	Channel Attention + Spatial Attention
Pooling Operation	MaxPooling2D
Upsampling Method	UpSampling2D
Feature Fusion	Concatenate Skip Connections
Output Layer	Conv2D (1 Channel Segmentation Output)
Total Parameters	1,975,502
Trainable Parameters	1,975,502
Non-Trainable Parameters	0



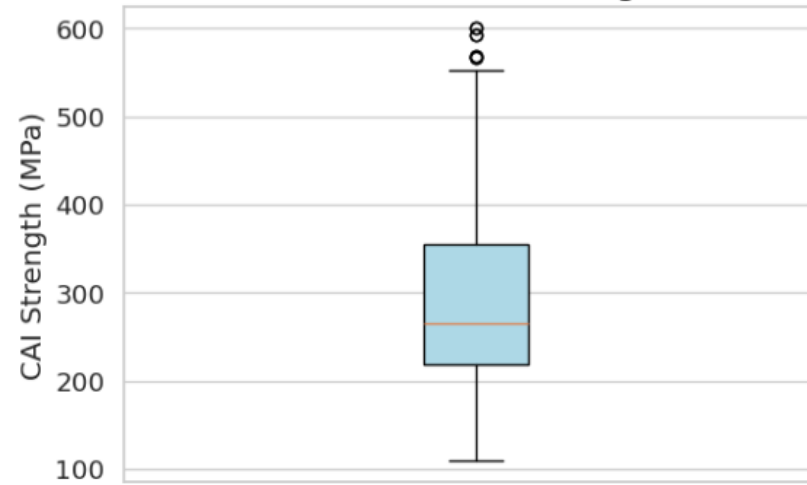
Epoch	Train Accuracy	Train Loss	Val Accuracy	Val Loss
1	0.8470	0.0926	0.8584	0.0923
5	0.8489	0.0881	0.8579	0.0927
10	0.8498	0.0861	0.8577	0.0922

CFRP CAI Strength — Exploratory Data Analysis

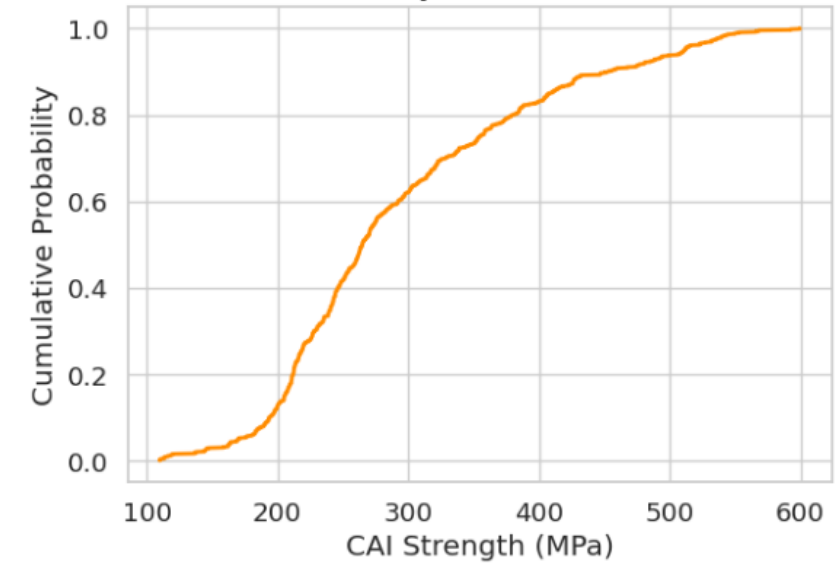
Distribution of CAI Strength



Box Plot of CAI Strength



Empirical CDF



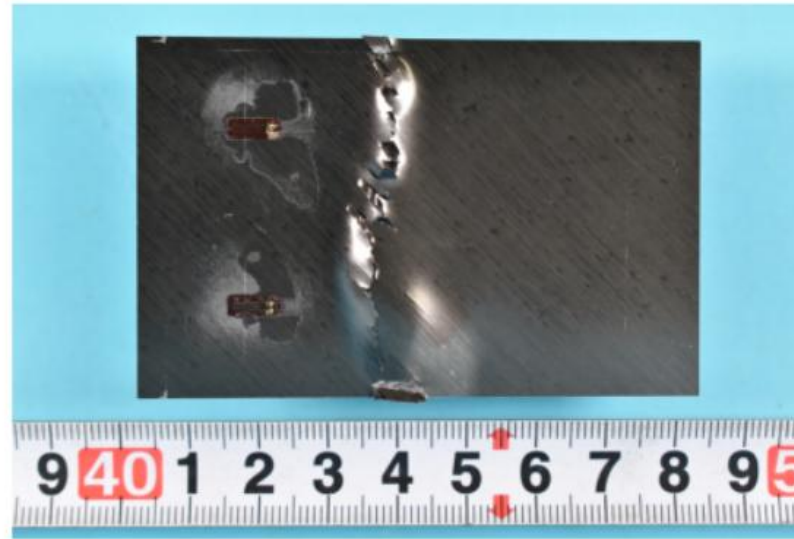
Strength stats (MPa):

Min = 109.75
Max = 599.57
Mean = 294.71
Std = 101.79
Count = 868

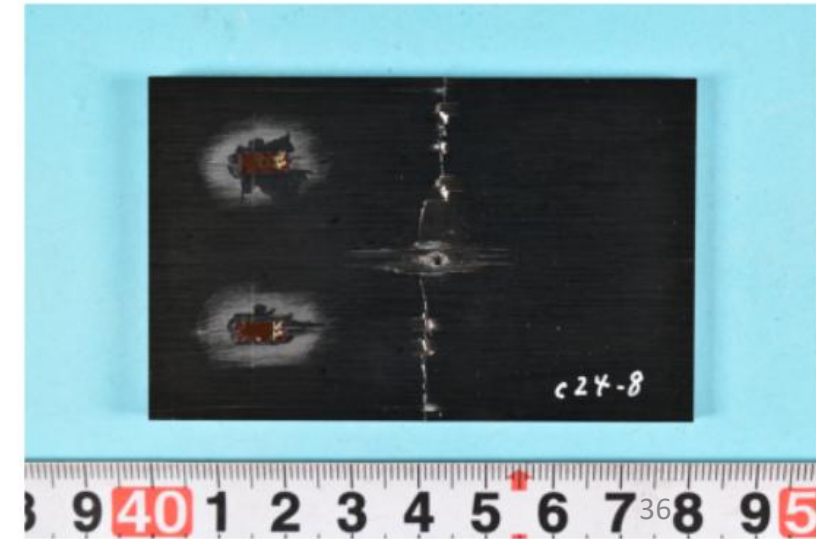
q8-12_back
204.9 MPa



q24-4_back
363.9 MPa



c24-8_front
319.4 MPa



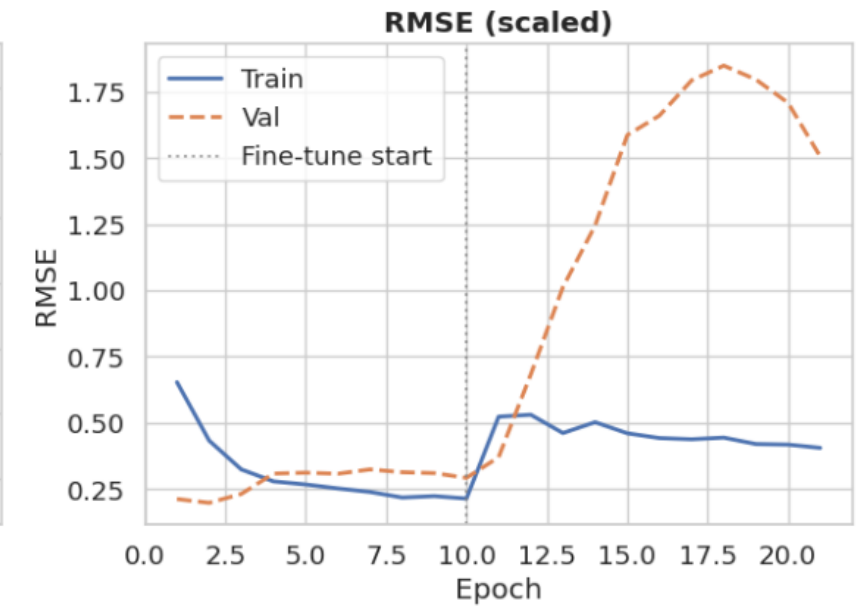
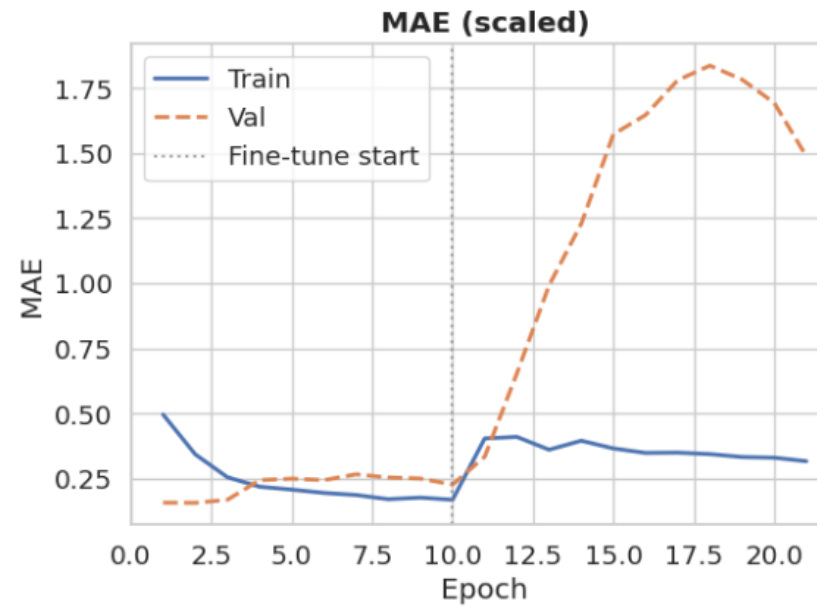
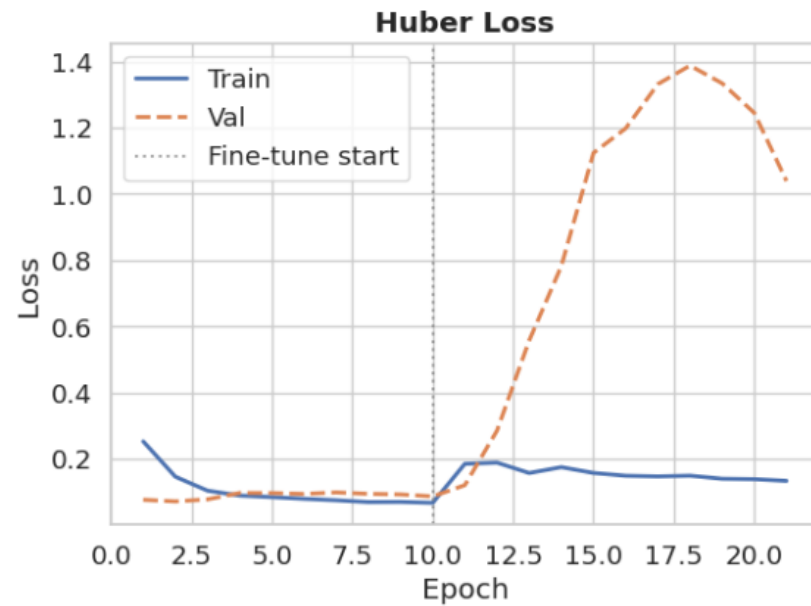


Results Branch C

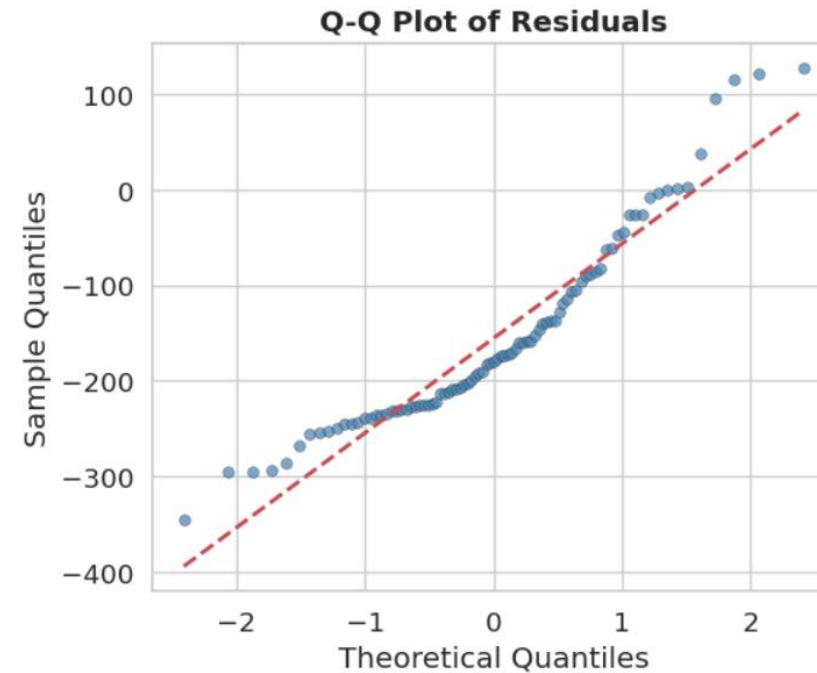
Train: 694 samples | $\mu=294.76$ MPa | $\sigma=102.99$ MPa
Val : 87 samples | $\mu=295.92$ MPa | $\sigma=96.25$ MPa
Test : 87 samples | $\mu=293.14$ MPa | $\sigma=99.18$ MPa

Layer / Component	Output Shape	Parameters
Input Layer	(224, 224, 3)	0
MobileNetV2 Backbone	(7, 7, 1280)	2,257,984
Global Average Pooling	(1280)	0
Batch Normalization	(1280)	5,120
Dense Layer (256 Units)	(256)	327,936
Dense Layer (128 Units)	(128)	32,896
Output Layer	(1)	129
Total Parameters	—	2,624,065
Trainable Parameters	—	363,521
Non-Trainable Parameters	—	2,260,544

Training History — Phase 1 + Phase 2

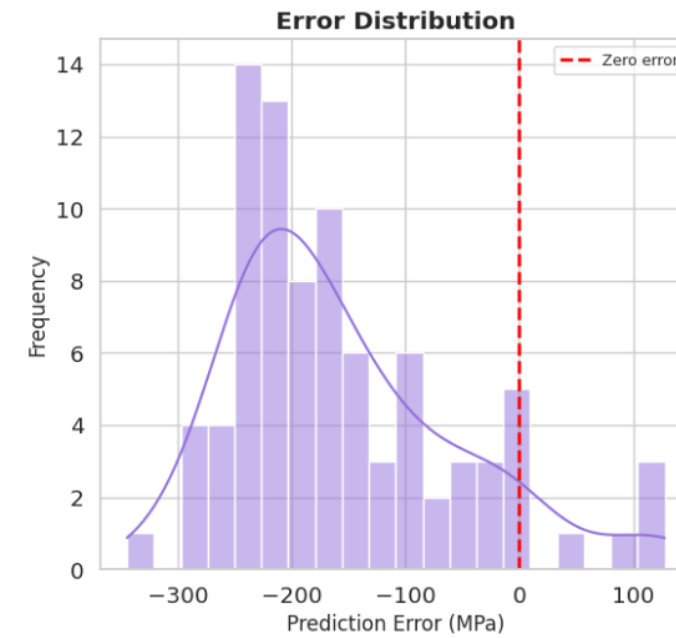
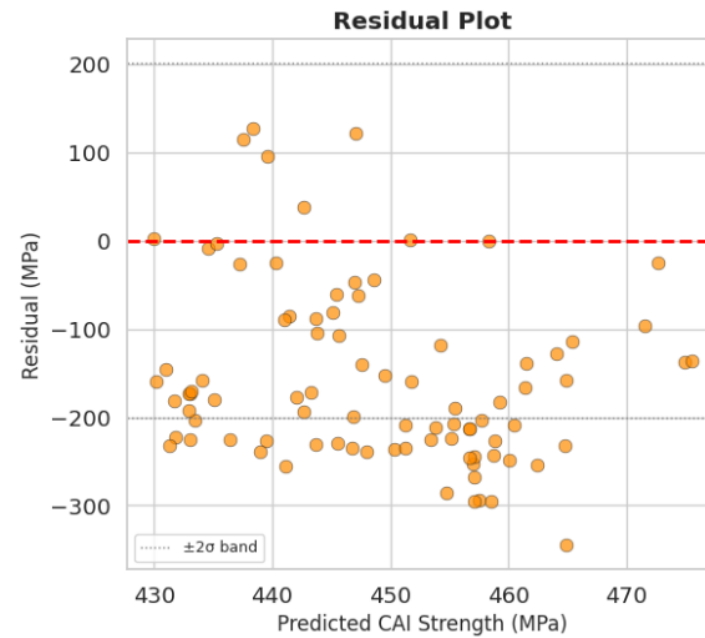
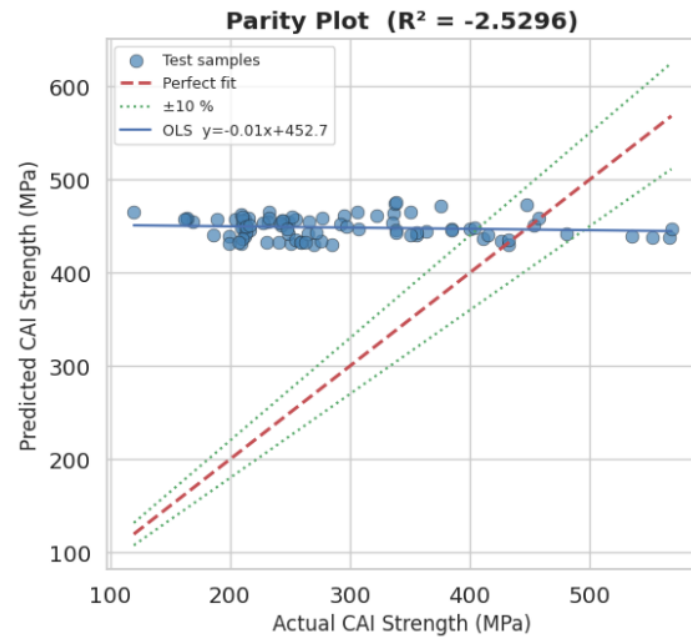


Test Set Performance Metric	Value
MAE	167.0958 MPa
RMSE	185.2612 MPa
NRMSE	41.31 %
MAPE	71.70 %
R ² Score	-2.5296



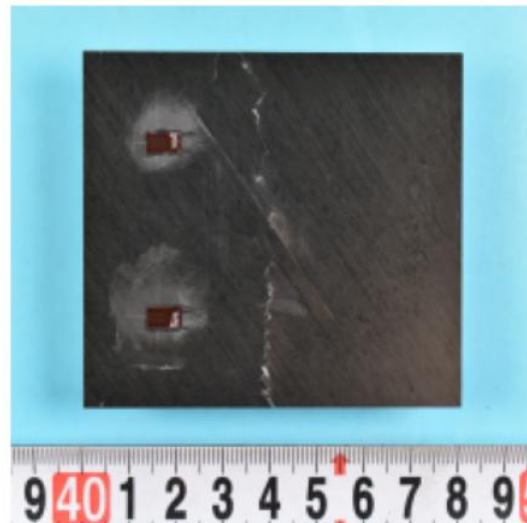
Shapiro-Wilk test: $W=0.9215$, $p=0.0001$
 → Residuals deviate from normality ($p \leq 0.05$)

CFRP CAI Strength Prediction — Evaluation Plots

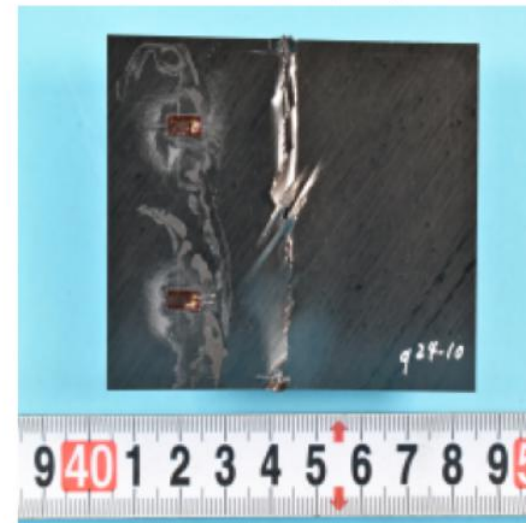


Prediction Examples on Test Samples

Actual: 213.9 MPa
Pred : 450.4 MPa
Error : 236.5 MPa (110.6%)



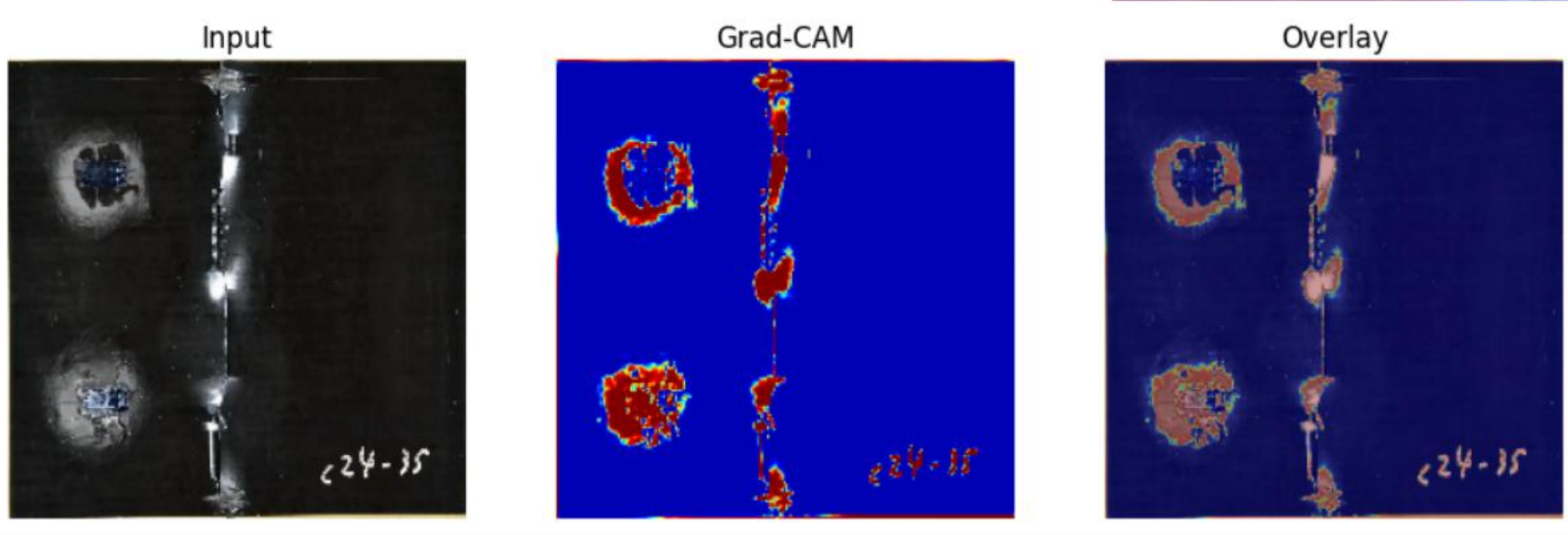
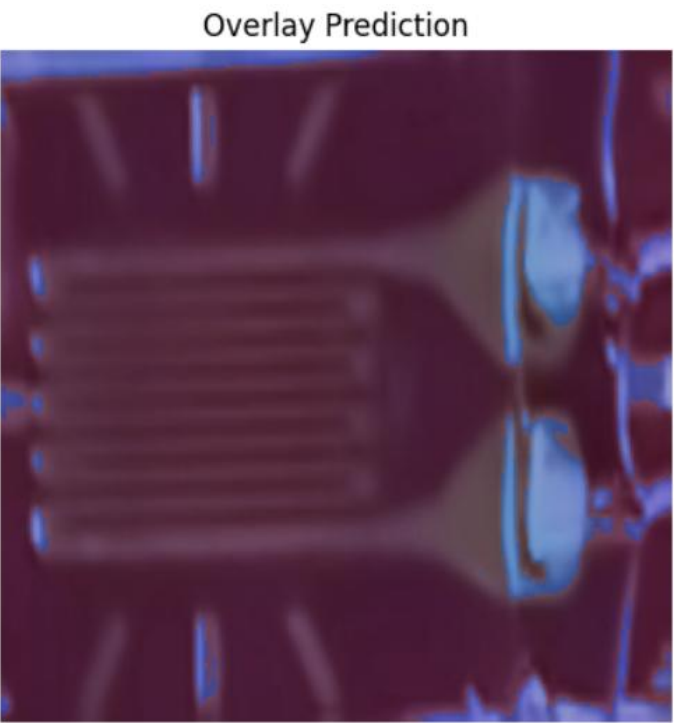
Actual: 336.4 MPa
Pred : 464.1 MPa
Error : 127.7 MPa (38.0%)



Actual: 350.8 MPa
Pred : 465.4 MPa
Error : 114.7 MPa (32.7%)



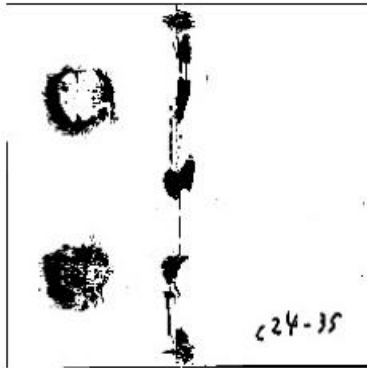
Evaluation Metric	Value	Description
Dice Coefficient	0.9761	Measures overlap accuracy between predicted and actual segmentation masks
IoU Score	0.9533	Evaluates intersection over union segmentation performance
Precision	0.9877	Measures correctness of predicted damaged regions
Recall	0.9837	Measures ability to detect actual damaged regions
F1-Score	0.9857	Harmonic mean of precision and recall
R ² Score	0.7556	Indicates overall prediction consistency and fitting quality



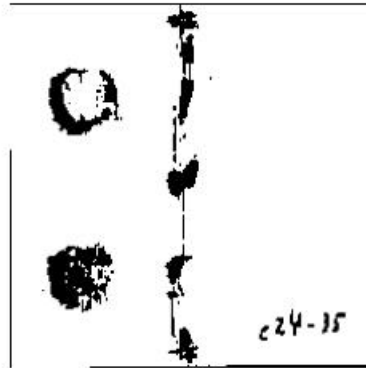
Input



GT



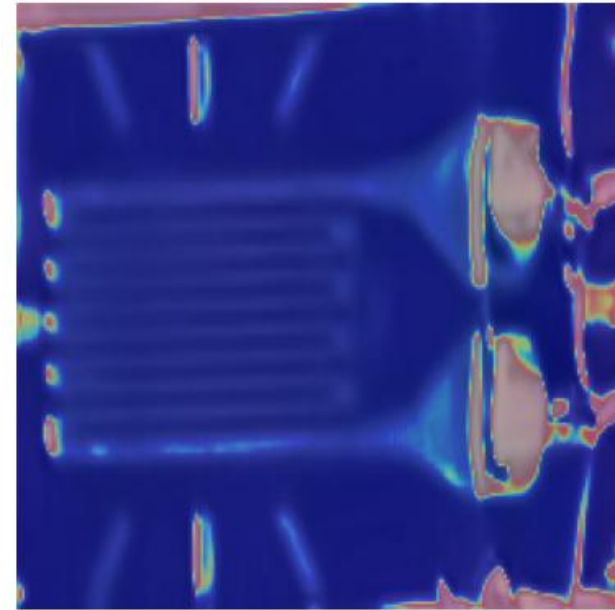
Pred



Sample 0



Sample 1



Input



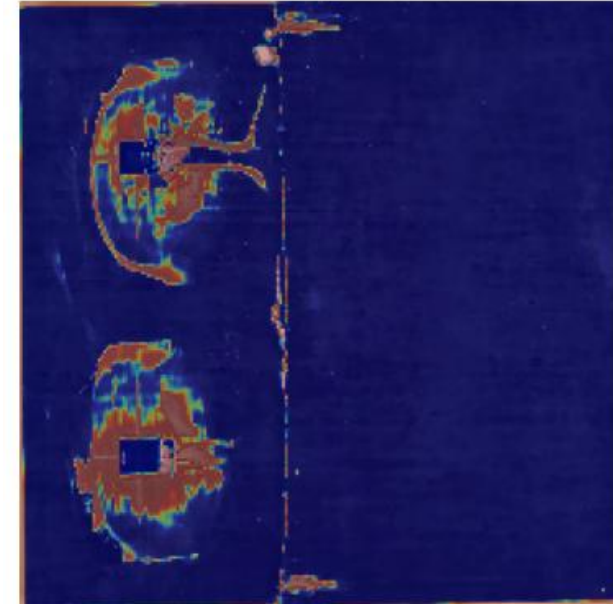
GT



Pred



Sample 2



Input



GT



Pred





Challenges

Complex CFRP Damage Patterns :Difficulty in accurately detecting cracks, delamination, and fibre breakage due to irregular and multi-scale damage structures.

Limited High-Quality Annotated Datasets: Availability of labeled CFRP segmentation and CAI datasets was limited for robust deep learning training.

Class Imbalance & Data Variability: Uneven distribution of damaged and pristine samples affected model generalization and prediction stability.

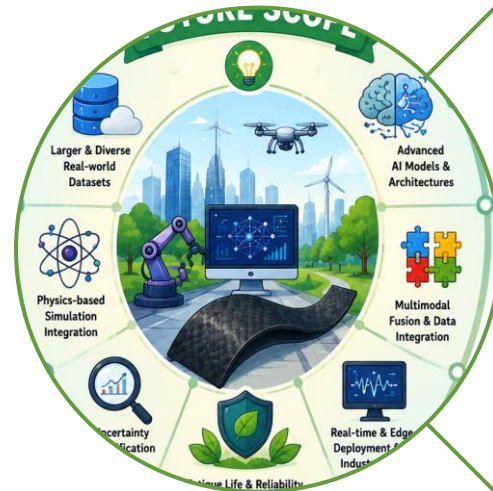
Noise and Background Artifacts in Images: CFRP images contained illumination variations, noise, and unwanted background regions that reduced segmentation accuracy.

Balancing Accuracy with Explainability: Integrating SHAP, LIME, and Grad-CAM while maintaining high predictive performance increased model complexity.

Conclusions & Future scope



Conclusion: This work presents a unified multimodal explainable deep learning framework for composite material analysis and damage assessment in CFRP structures. The proposed system achieved strong performance in strength prediction, structural integrity classification, damage segmentation, and CAI strength regression using DNN Ensemble, MaterialsDNN, Hybrid Attention U-Net, and MobileNetV2 architectures.



Future Scope: Future work will focus on multimodal fusion using larger real-world CFRP datasets and integration of physics-based simulations for improved generalization. Further improvements include lightweight edge deployment, uncertainty quantification, fatigue life prediction, and advanced explainable AI techniques for real-time structural health monitoring applications.



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Thank You!

Thank you for your time
and attention!

A Unified Multimodal Explainable Deep Learning
Framework for Composite Material Analysis
and Damage Assessment in CFRP Structures

